Cyber Deception: Taxonomy, State of the Art, Frameworks, Trends, and Open Challenges

Pedro Beltrán-López, Manuel Gil Pérez, and Pantaleone Nespoli

***Abstract*—The growing interest in cybersecurity has significantly increased articles designing and implementing various Cyber Deception (CYDEC) mechanisms. This trend reflects the urgent need for new strategies to address cyber threats effectively. Since its emergence, CYDEC has established itself as an innovative defense against attackers, thanks to its proactive and reactive capabilities, finding applications in numerous real-life scenarios. Despite the considerable work devoted to CYDEC, the literature still presents significant gaps. In particular, there has not been (i) a comprehensive analysis of the main components characterizing CYDEC, (ii) a generic classification covering all types of solutions, nor (iii) a survey of the current state of the literature in various contexts. This article aims to fill these gaps through a detailed review of the main features that comprise CYDEC, developing a comprehensive classification taxonomy. In addition, the different frameworks used to generate CYDEC are reviewed, presenting a more comprehensive one. Existing solutions in the literature using CYDEC, both without Artificial Intelligence (AI) and with AI, are studied and compared. Finally, the most salient trends of the current state of the art are discussed, offering a list of pending challenges for future research.**

***Index Terms*—Cybersecurity, cyber deception, honeypot, survey, active defense, moving target defense, large language model, federated learning, artificial intelligence.**

I. Introduction

EMERGING in the contemporary digital age, organizations of all sectors face a growing threat of cyberattacks that compromise the confidentiality, integrity and availability of their data and systems [1], [2]. These attacks, ranging from the theft of personal information to the sabotage of critical infrastructures [3], have become increasingly sophisticated and difficult to detect and respond. In this context, cybersecurity has become a strategic priority for businesses and governments [4].

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Nevertheless, traditional security measures are often insufficient to protect against Advanced Persistent Threats (APT) [5] or zero-day attacks [6]. This critical factor highlights the need for novel and effective defense strategies capable of preempting these attacks and mitigating them more effectively [7]. This is where CYDEC comes into play, an innovative strategy redefining the cybersecurity landscape.

CYDEC [8] is based on the principle of deception [9], a strategy used in various forms of conflict throughout history, from warfare to espionage. This form of defense makes it possible to fight attacks preemptively and reactively [10] to mitigate them efficiently and without consequences. In doing so, it not only diverts the attacker's attention but also gathers valuable intelligence on their methods, tools, and targets, creating a context of continuous learning [11].

CYDEC offers multiple advantages. In particular, it facilitates the detection of threats that conventional measures may miss by collecting information obtained from deception [12]. Any interaction with these decoys is automatically identified as suspicious, enabling faster and more accurate detection. In addition, by interacting with the decoys, attackers reveal Tactics, Techniques, and Procedures (TTPs) [13] that can be used to strengthen defenses and anticipate future attacks. Understanding and predicting attacker behavior allows organizations to adjust their security strategies proactively [14]. CYDEC also wears down and deters attackers, creating a more complex and challenging environment, increasing operational costs, and decreasing the likelihood of success [15]. Finally, it is a cost-effective solution, as creating and maintaining decoys is often less expensive than protecting every asset in an organization against all possible threats [16].

Although the CYDEC has been an integral part of our defenses for some time, it is in recent years that its value has been significantly enhanced due to the numerous benefits it offers. As depicted in Fig. 1, since 2018, articles in the literature have been on the rise due to increased attention from both industry and academia.

A. Existing CYDEC Survey

Therefore, studies have been conducted that compile and analyze existing research on CYDEC, as shown in Table I. These surveys have been analyzed because of their great impact on the scientific community, since previous surveys are too old and do not include the main characteristics analyzed. The characteristics analyzed for each of the surveys are as follows: Fundamentals (the survey describes the fundamental

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TABLE I
COMPARISON OF SURVEYS ANALYZING CYDEC

Ref	Year	Papers analyzed	Fundamentals	Taxonomy Proposal	Framework	Attack	AI	TRL	Open challenges
[18]	2018	51	✓	✓	✗	✗	✗	✗	✗
[19]	2020	24	✗	✓	✗	✗	✗	✗	✗
[20]	2021	*N/S*	✗	✓	✗	✗	✗	✗	✗
[21]	2021	56	✓	✓	✗	✗	?	✗	✓
[22]	2021	27	✓	✗	✗	✗	✗	✗	✓
[23]	2022	77	✗	✗	✗	✗	?	✗	✓
[24]	2024	73	?	✗	✗	✗	✗	✗	✓
Ours	2024	95	✓	✓	✓	✓	✓	✓	✓

N/S (Not Specified) by the authors, ✓ addressed, ✗ not addressed by the work, ? partial addressed by the work

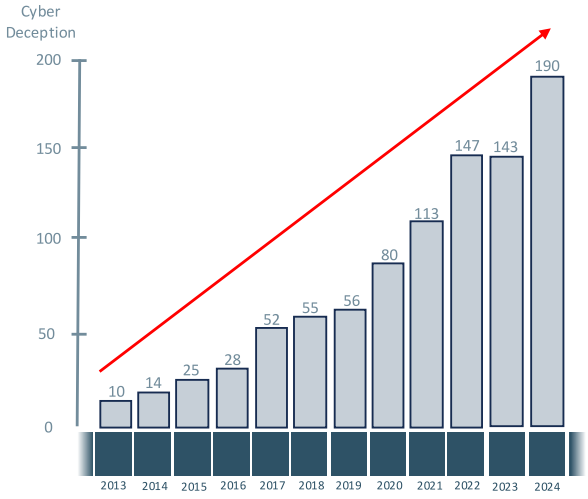


Fig. 1. CYDEC Article Timeline (source: Web of Science [17]).

aspects of CYDEC), Taxonomy Proposal (the survey proposes its own taxonomy), Framework (the survey analyzes articles based on frameworks), Attack (the survey analyzes articles based on attack approach), Artificial Intelligence (AI) (the survey analyzes deception mechanisms using different AI modalities), Technological Readiness Level (TRL) (the survey analyzes the TRL of the deception mechanisms analyzed) and Open challenges (the survey presents open challenges).

A comprehensive review in [18] categorized various methodologies and tools available, providing a detailed overview of decoy and honeypot techniques. The research emphasized the effectiveness of these techniques in confusing and trapping attackers, in addition to gathering valuable information about their tactics and methods.

Exploring a different point of view, article in [19] presented how game theory can be used to model strategic interactions between attackers and defenders. The authors developed a detailed taxonomy organizing various defensive strategies based on game theory principles, highlighting the importance of anticipating and countering attackers'.

Providing a historical perspective, [20] offered a retrospective and prospective view of deception techniques in active cyber defense. The authors reviewed the evolution of these techniques over the past three decades, identifying key milestones and important developments. They projected future trends and challenges, underscoring the need to innovate and evolve deception strategies to address emerging cyber threats.

Delving into the combination of different disciplines, the study in [21] reviewed defensive deception techniques that integrated game theory and Machine Learning (ML). This article highlighted how these two disciplines can come together to create more robust and adaptive strategies, illustrating with case studies the practical application of these techniques in cyberattack detection and mitigation. The authors concluded that this integration provided a powerful tool for cyber defense, enhancing the ability of organizations.

Additionally, [22] analyzed the challenges and opportunities of implementing deception techniques. Despite their potential, adoption of these techniques has been limited due to the high costs and complexity of generating realistic deception artifacts. However, recent advances in ML offered new possibilities for automating and scaling the creation of these artifacts, proposing a promising future for CYDEC and underscoring the need to develop models that could mimic common components of the Information Technology (IT) environment convincingly.

In addition [23] reviewed the use of computational intelligence (ML and Deep Learning (DL)) techniques for defensive deception leaving RF techniques, for example, unanalyzed. They discussed recent advances and open problems, highlighting how AI techniques could improve deception creation and management. The article discussed practical applications and case studies that demonstrated these techniques' effectiveness, identifying future areas of research.

Last but not least, a comprehensive survey in [24] reviewed CYDEC techniques for enhancing honeypot performance. It emphasized how honeypots, as decoy systems, confuse and delay attackers while collecting critical intelligence on their behavior. The research systematically classified deception techniques and evaluated their effectiveness through simulations and mathematical modeling. Notably, it identified gaps such as the need to adapt honeypots to 5G and Software Defined Networking (SDN) vulnerabilities, projecting the evolution of deception strategies. The authors concluded with a roadmap for future research, highlighting the importance of continual innovation.

B. Research Questions

This article addresses the identified gaps by analyzing the different existing frameworks on CYDEC, and proposes a conceptual framework that incorporates AI and the different

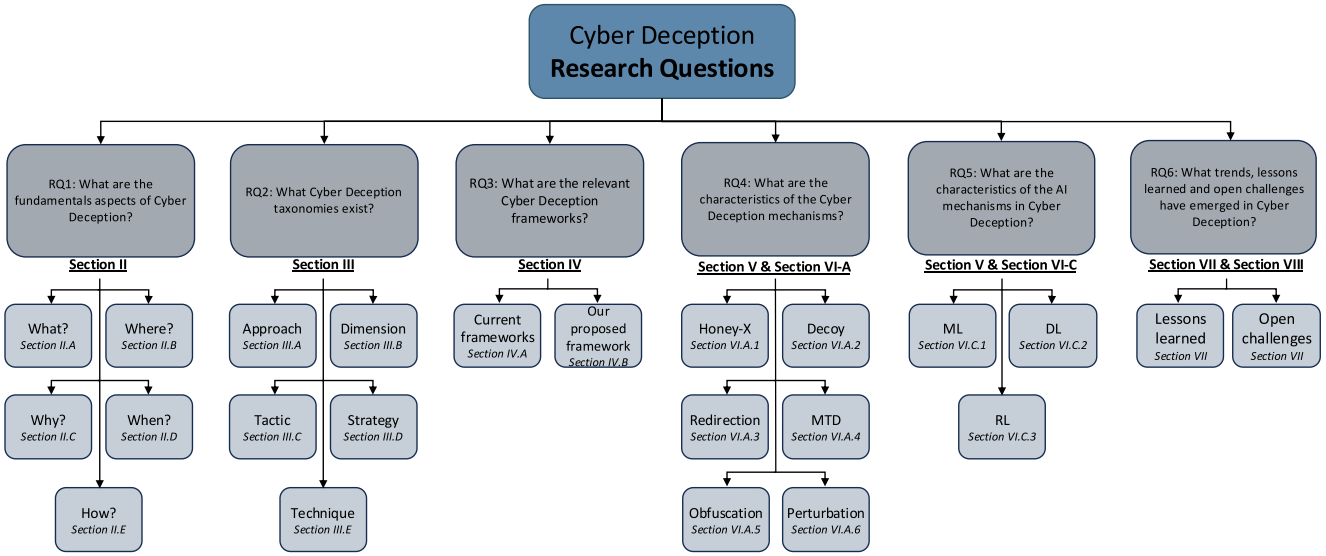


Fig. 2. Research questions and sections answering them.

tactics of defense. In addition, a robust and comprehensive taxonomy, capable of comprehensively classifying each mechanism of CYDEC, is also presented. In addition, the current articles on CYDEC in the literature and the different mechanisms of AI existing in this field are analyzed using different metrics that no other survey has used such as TRL, Attack, Dimensions or Tactics, among others.

Within this context, and adhering to the storyline outlined in Section I, it becomes essential to address the following Research Questions (RQ):

- *RQ1: What are the fundamentals aspects of CYDEC?* This question delves into the core principles and foundational concepts of CYDEC. It aims to uncover the essential elements that make deception a viable and effective strategy.
- *RQ2: What CYDEC taxonomies exist?* By examining existing taxonomies, this question assesses their effectiveness in categorizing various deception mechanisms. It looks into how these taxonomies help organize and understand the diverse strategies employed in CYDEC.
- *RQ3: What are the relevant CYDEC frameworks?* This inquiry surveys the landscape of CYDEC frameworks, identifying their key characteristics and methodologies. It seeks to highlight the distinct features and structures that define these frameworks.
- *RQ4: What are the characteristics of the CYDEC mechanisms?* This question investigates the essential traits of highly impactful CYDEC mechanisms used in defense and offense. It focuses on these mechanisms' operational aspects and effectiveness in various scenarios.
- *RQ5: What are the characteristics of the AI mechanisms in CYDEC?* Exploring the integration of AI in CYDEC, this question identifies the primary features and capabilities of AI-driven techniques. It looks at the role of AI in enhancing deception strategies and its impact.
- *RQ6: What trends, lessons learned and open challenges have emerged in CYDEC?* This inquiry aims to uncover current trends and emerging challenges within the field of CYDEC. It examines recent advancements and highlights the unresolved issues that present future research and innovation opportunities.

C. Contributions

Observing Table I, it is possible to note that, although different studies have been carried out that compile the state of the art in the field of CYDEC, the most recent studies primarily focus on the analysis of Game Theory or the mechanisms of CYDEC employing ML and DL, often without delving into a comprehensive exploration of CYDEC across all its AI dimensions. Unlike these prior works, our paper presents a holistic review that not only synthesizes existing CYDEC mechanisms but also analyzes their integration with diverse AI techniques, providing a broader and deeper understanding of the domain. Furthermore, while some existing surveys propose taxonomies, we identify significant gaps in their comprehensiveness and robustness. In this paper, we introduce a novel taxonomy designed to rigorously classify CYDEC mechanisms, offering a more systematic and insightful perspective. Additionally, none of the reviewed surveys analyze CYDEC frameworks in detail, nor do they propose a unified framework that integrates the benefits of AI. Addressing this gap, our work not only critiques existing frameworks but also proposes a new framework that leverages AI to enhance the effectiveness of CYDEC.

Fig. 2 shows where and how the above questions are addressed in this document as a table of contents. Specifically, Section II discusses the fundamentals of CYDEC with emphasis on the five fundamental questions (what, where, why, when, and how). Section III reviews the main taxonomies in the literature, identifying their advantages and disadvantages. Subsequently, our taxonomy is presented, capable of comprehensively classifying any CYDEC mechanism. Section IV discusses the CYDEC frameworks identified in the taxonomy, emphasizing the tactics they cover, whether they use AI, the dimensions, or the application scenario. In addition, Section IV

presents our CYDEC framework capable of preventing, detecting, reacting, and generating forensic analysis with the help of AI. Next, Section VI describes and compares the main CYDEC solutions found in the state-of-the-art, analyzing their main features identified in Section V. Subsequently, Section VI-C performs the same analysis of the previous section but pays special attention to the use of AI in the CYDEC field. Section VIII presents a set of future research challenges. Finally, Section IX provides an overview of the conclusions drawn from the work.

II. Background

This section provides a comprehensive definition of CYDEC as recognized by the scientific community. It aims to thoroughly specify the primary principles of CYDEC to address *RQ1* (refer to Fig. 2). To fully elucidate CYDEC, five fundamental inquires are addressed throughout this section: what CYDEC is, where CYDEC is used, why CYDEC is used, when CYDEC is used, and how CYDEC is used. The *what* clarifies its theoretical basis; the *where* situates its domain; the *why* exposes its benefits; the *when* pinpoints optimal timing; and the *how* details practical deployment steps.

A. What?

CYDEC [25], also known as active defense, is an advanced computer security strategy that uses different mechanisms to make a specific target believe something it is not. Gene Spafford first introduced this concept in 1989 [26]. Gene referred to an active defense to identify attacks that have already begun, identify the attackers' techniques, and provide them with false data.

It is worth noting that CYDEC is a strategy that goes beyond simply using decoys. For instance, it involves creating dynamic and realistic deception strategies tailored to confuse and mislead adversaries [27]. A financial institution might implement CYDEC by deploying a realistic but isolated decoy database containing fake customer data. This setup can identify adversaries attempting to exfiltrate information while safeguarding real customer data. This foundational approach differentiates CYDEC from passive defenses, emphasizing active manipulation rather than static barriers.

Concretely, CYDEC can be used for both attack and defense [28]. In the case of defense, defenders can carry out actions that deceive the attacker to mitigate and even learn from his/her actions. In the case of an attack, attackers can use deception to improve stealth when entering a system or make the defender focus on some hidden point in the system.

In essence, CYDEC is based on the premise that defenders and attackers can improve their security posture not only by fortifying their systems and networks but also by actively manipulating the digital environment to confuse, disorient, and frustrate [29], [30].

B. Where?

The versatility of CYDEC, coupled with its adaptability, renders it applicable across a spectrum of defensive and offensive scenarios [31]. In defensive operations, CYDEC can be a robust shield against malicious actors, employing deceptive strategies to mislead and thwart adversaries' advances. CYDEC is applied across various environments, tailored to specific operational contexts. For instance, critical infrastructures like power grid control systems can deploy decoy Industrial Control System (ICS) that mimic real operations. In information protection, honeytokens placed in sensitive file directories mislead attackers and signal unauthorized access attempts.

On the offensive front, CYDEC becomes a potent weapon, harnessing the power of deception to outmaneuver opponents and gain strategic advantage. By obscuring true intentions, manipulating perceptions, and exploiting vulnerabilities, CYDEC enables proactive engagement while avoiding detection.

C. Why?

CYDEC has emerged as a cornerstone in organizations' cyber defense toolkit, and this prominence is justified for several key reasons that reflect its relevance and effectiveness in today's cybersecurity landscape [32], [33]. In addition, CYDEC's flexibility to adapt to diverse environments and scenarios, as well as its ability to generate intelligence on adversaries' Tactics, Techniques, and Procedures (TTP), are aspects that underscore its strategic importance in protecting critical assets and mitigating threats. The following are some of the advantages of using CYDEC:

- Making it more difficult for attackers: CYDEC introduces uncertainty and confusion into the cyber environment, making the attackers' task more difficult [34]. Flooding the digital space with lures and traps increases complexity for adversaries. For instance, an e-commerce company might deploy decoy user accounts with fake transaction histories.
- Improve early detection: By placing digital decoys at strategic locations in the network, CYDEC facilitates early detection of malicious activity. These false signals act as early warning signs, allowing security teams to quickly identify and respond to intrusion attempts before they can cause significant damage [35].
- Reduce attack surface: By implementing CYDEC measures, organizations can effectively reduce their attack surface, decreasing the opportunities for attackers to find and exploit vulnerabilities in their systems and networks [36].
- Discourage adversaries: The presence of CYDEC elements can deter cyber adversaries by leading them to believe that the environment is well protected and that their activities can be easily detected and thwarted. This heightened risk perception can lead attackers to seek easier and less protected targets rather than confront robust defenses [37].
- Gather Threat Intelligence: Cyber deception protects against cyberattacks and provides a unique opportunity to gather intelligence on adversaries' TTP [38]. By observing how attackers interact with lures and traps, security teams can gain valuable information to improve their

defense strategies further and anticipate future attacks by reducing the impact of 0-days or detecting behaviors characteristic of these attacks.

D. When?

In the attack context, CYDEC offers tools and methodologies that enable attackers to enhance their stealth strategies and deter adversary defenses. These strategies can be found throughout the lifecycle of an attack defined by attackers knowledge bases such as MITRE ATT&CK [39], from the reconnaissance and entry tactic through execution to maintaining access and data exfiltration. CYDEC facilitates both the identification of vulnerabilities in systems and networks and the planning and execution of effective attacks.

On the defensive side, CYDEC provides resources to harden infrastructures and protect them against intrusions and exploits and can be used for prevention, detection, reaction, or forensic analysis. Timing is a critical factor in CYDEC strategies, as it determines when deception strategies should be deployed for maximum effectiveness. For example, during the reconnaissance tactic of a cyberattack, deploying deceptive DNS records can mislead attackers about the organization's actual infrastructure. Similarly, in the detection tactic, decoy credentials can trigger alerts when misused. In later stages, such as during exfiltration attempts, decoy files may delay or thwart the adversary's efforts [40].

In addition, the versatility of CYDEC allows it to be applied in both real-time situations, e.g., to respond immediately to emerging threats, and in non-real-time contexts, e.g., to perform retrospective analysis and continuously improve cybersecurity. This means that organizations can use CYDEC not only as a reactive tool to address current threats but also as a strategic platform to strengthen their defenses and anticipate future challenges in the cybersecurity landscape.

E. How?

Effective implementation of CYDEC involves combining techniques and tools designed to deceive cyber adversaries and strengthen an organization's defenses [41]. There are various ways to carry out deception actions depending on whether to attack or defend. This discipline encompasses a variety of creative approaches to deceive adversaries, confusing and discouraging them from their malicious objectives. While there is no single way to conduct CYDEC, its successful implementation involves thoroughly understanding the strategies and techniques available and the organization's specific infrastructure and environment. For instance, deploying Moving Target Defense (MTD) tools in cloud environments can rotate IP addresses to confuse attackers and reduce exposure. Similarly, open-source platforms like Honeyd can simulate entire networks to mislead adversaries. The success of CYDEC depends on understanding the organization's unique infrastructure and selecting tools that align with its specific cybersecurity objectives. Numerous tools simplify the way CYDEC techniques are achieved, as will be shown in Sections III-D and III-E.

III. Taxonomy

Currently, several taxonomies have been proposed to classify the different CYDEC techniques according to specific criteria, both in terms of attack [42] and defense [19], [43], [44]. However, none of these classifications has considered such important aspects as the dimension in which deception is employed or whether it is used to attack or defend. Moreover, another crucial point that differentiates our taxonomy from previous proposals lies in classifying deception according to the tactic in which they are employed and include more CYDEC techniques. Considering the limitations of previous taxonomies, in Fig. 3 a detailed taxonomy is presented. With this proposed taxonomy, the research question *RQ2* (refer to Fig. 2) is solved with this taxonomy, which focused on whether a complete taxonomy correctly classifies CYDEC mechanisms and, if not, propose one to classify them.

This taxonomy focuses on accurately characterizing the CYDEC technique used in each case. Moreover, it is important to note that we have divided the proposed taxonomy into five components, i.e., Approach, Dimension, Tactic, Strategy, and Technique. With these components, we can characterize any deception mechanism present in the current literature. Furthermore, the main characteristics of each deception mechanism used can be described and specified in great detail.

A. Approach: Classifying Deception as Defensive or Offensive

The first step in characterizing the technique is identifying whether it is a defensive or an attacking action.

- Attack [45]: Attack by deception is a sophisticated cybercriminal approach to compromise systems and access sensitive information. It involves manipulating data, impersonation, digital decoys, and disguised malware. These attacks are difficult to detect and mitigate as they aim to confuse and evade defenses. For example, an attacker may inject false information or execute a Man-In-The-Middle (MITM) attack.
- Defense [46]: Defense by deception is a proactive or reactive strategy to counter cyber threats by confusing attackers. It employs decoys, honeypots, and false data to divert adversaries from real assets, increasing detection chances and reducing breach success.

B. Dimension: Application Domains of Cyber Deception

The dimension is intended to identify the component(s) used to deceive the attacker or defender [47].

- Data [48]: Data manipulation aims to erode adversaries' trust in information, weakening their strategies. It involves altering authentication, system activity, or information flow, either by falsifying records or subtly modifying data.
- Network [49]: Network deception manipulates transmitted data and components through redirections, decoys, or modifications. Attackers use it to mask activities, while defenders leverage it to detect threats and mislead adversaries.

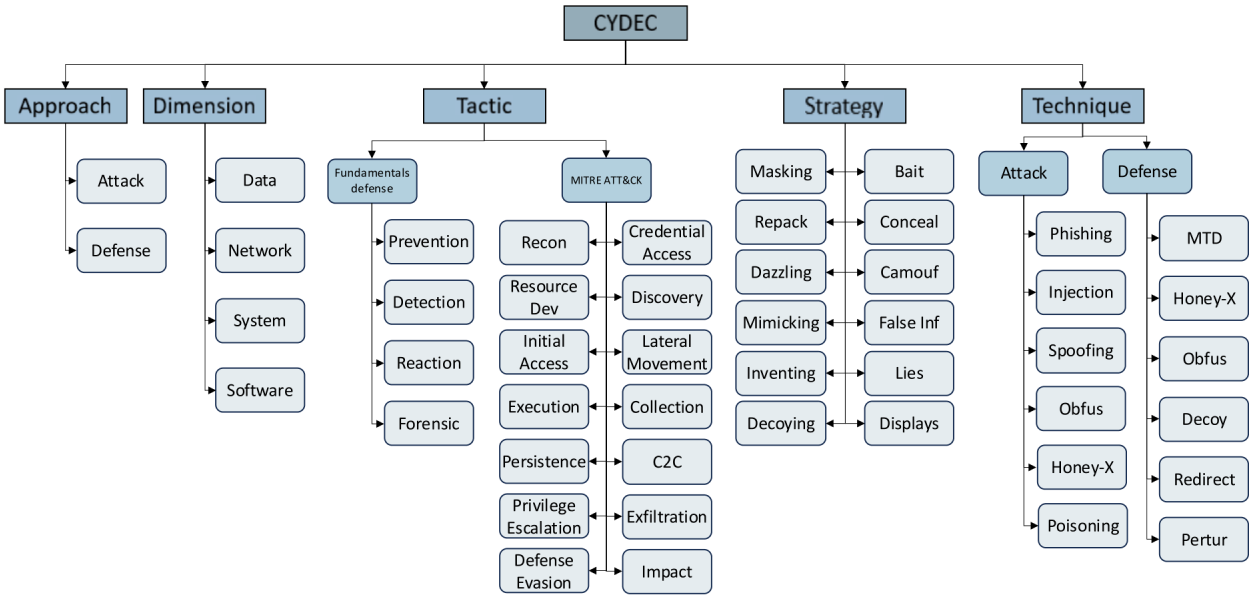


Fig. 3. Proposed taxonomy of CYDEC mechanisms organized by levels.

- System [50]: This dimension alters essential system components, including OS, hardware, and storage, to deceive adversaries. Attackers introduce vulnerabilities or suspicious behavior, while defenders create traps to expose malicious intent.
- Software [18]: Software deception manipulates applications and scripts. Attackers inject malicious code to exploit vulnerabilities, whereas defenders create fake applications to lure and neutralize threats.

C. Tactic: Positioning Deception Across Attack and Defense Lifecycles

The tactic focus on positioning the deception mechanism used at a particular point in the defense or attack's life cycle.

- MITRE ATT&CK [51]: MITRE ATT&CK tactics categorize adversary behaviors, enhancing defense strategies by aligning CYDEC solutions with the 14 attack lifecycle stages.
- Fundamentals defense tactics [52]: To represent the tactics of defense, the four fundamental tactics have been used, namely, prevention, detection, reaction, and forensics. These tactics provide a solid framework for understanding how to approach cybersecurity challenges from different perspectives. The same approach has not been followed as with the attack using MITRE ATT&CK, because MITRE Defend [53] is at a very early stage of development, and we do not consider it to be fully aligned with the functions of our taxonomy. Moreover, in MITRE Defend, deception is only present in one of the tactics of the defense and, in our humble opinion, deception should be present in all tactics of the defense as another component that can be used to reinforce security. Additionally, MITRE Engage [54], which focuses on adversary engagement operations, has not been used for similar reasons. MITRE Engage is also in its nascent stages and lacks the maturity and comprehensive integration needed to fit seamlessly into our existing taxonomy. Moreover, the concept of adversary engagement, like deception, is seen as an overarching strategy rather than a distinct tactic.

D. Strategy: Goals of Deception

There are several ways to carry out a deception. Therefore, characterizing the purpose of the deception mechanism, i.e., what objective or objectives at the level of deception are intended to be achieved, it is necessary to describe it. To carry out this characterization, we will use some strategies used by some of the surveys analyzed in the state of the art that have been obtained from [55], [56], [57] studies.

In addition, each of the techniques and strategies subsequently defined will be accompanied by an icon used later in Section VI.

MASKING [55], [56]: Hides sensitive information to prevent detection by adversaries. It is highly scalable due to encryption and service obfuscation with minimal resource overhead. For example, encrypting database query logs can prevent attackers from identifying access patterns, while services can be hidden by operating on non-standard ports.

REPACKAGING [55], [56]: Alters the appearance of digital objects to evade detection or exploit user trust. Effective for targeted attacks but resource-intensive at scale. An example is disguising critical files as innocuous images or text documents to deter attackers.

DAZZLING [55], [56]: Overwhelms adversaries with excessive information, making analysis difficult. Highly scalable through automation. For instance, defenders can flood an attacker's monitoring system with fake alerts or dummy network traffic, delaying their ability to identify real threats.

MIMICKING [55], [56]: Replicates legitimate entities or behaviors to deceive attackers. Scalable with automation,

though high-fidelity mimicking requires significant resources. A common use case is deploying fake login pages in honeypots to capture credentials.

INVENTING [55], [56]: Creates fictitious elements to mislead adversaries. Highly scalable with automated tools for generating decoys and false data. An example is deploying fake administrative accounts to lure attackers into attempting privilege escalation.

DECOYING [55], [56]: Redirects adversaries to false targets to protect critical assets. Moderately scalable, depending on infrastructure complexity. For example, fake servers mimicking financial databases can divert attackers from real systems.

BAIT [55], [56]: Uses enticing but false information to deceive adversaries. Moderately scalable, requiring regular updates for effectiveness. An example is deploying files labeled *Confidential_Passwords* that trigger security alerts when accessed.

CONCEALMENT [57]: Actively hides critical resources or information. Highly scalable through encryption and network segmentation. For example, encrypting sensitive data and storing it in obscure file locations makes it difficult for attackers to find.

CAMOUFLAGE [57]: Disguises digital elements to blend into their environment. Moderately scalable, requiring precise adjustments to avoid detection. A use case includes embedding critical network traffic within normal traffic patterns to prevent identification.

FALSE INFORMATION [57]: Distributes misleading content to confuse adversaries. Highly scalable, as decoy information can be replicated easily. For instance, defenders can inject false configuration files with fake IP addresses to waste an attacker's time.

LIES [57]: Provides false responses to mislead attackers. Moderately scalable, requiring context-specific tailoring. An example is reporting incorrect software versions to deceive attackers into exploiting nonexistent vulnerabilities.

DISPLAYS [57]: Manipulates the presentation of information to mislead adversaries. Moderately scalable, depending on the complexity of deception. For example, a system might display misleading login error messages to trick brute-force attackers while logging their activity.

E. Technique: Specific Deception Methods for Attack and Defense

Finally, the technique or techniques used for the proposed deception, defined as specific methods, procedures or approaches applied to achieve the intended deceptive result, must be specified. These techniques are divided into defense and attack because there are several techniques that are independent depending on whether defense or attack is required.

F. ATTACK

PHISHING [58]: An attack where adversaries impersonate legitimate entities to steal sensitive information. Highly scalable due to automation, phishing campaigns leverage email spoofing and fake websites. For example, an attacker might send a bank-like email prompting users to enter credentials on a fraudulent login page.

INJECTION [59]: Inserts malicious code into applications or systems to exploit vulnerabilities. Moderately scalable but effective with automated tools. A common example is SQL injection, where attackers manipulate input fields to extract unauthorized data.

SPOOFING [60]: Falsifies identity or source information to gain trust or unauthorized access. Highly scalable with automation, attackers use techniques like email or IP spoofing. An example is forging an email sender to trick users into executing malicious actions.

OBFUSCATION [61]: Modifies code to evade detection and analysis. Moderately scalable, leveraging encoding or polymorphic techniques. For instance, attackers can encode scripts in base64 to bypass security filters.

HONEY-X [61]: Deploys decoys to mislead defenders. Highly scalable, depending on the realism of the decoys. Attackers may plant fake credentials in a system to divert security efforts while targeting valuable assets.

POISONING [62]: Manipulates data or systems to degrade operations or create vulnerabilities. Moderately scalable with automated tools. A common case is data poisoning in ML models, where injected false data skews predictive accuracy.

G. DEFENSE

MTD [63]: Dynamically reconfigures network assets to disrupt attacks. Highly scalable and effective with automation. For example, periodic IP and port changes prevent adversaries from maintaining persistent access.

HONEY-X [64]: Uses decoys and traps to detect and analyze attacks. Moderately scalable with proper planning. A common example is a fake admin portal that logs unauthorized access attempts.

OBFUSCATION [65]: Confuses adversaries by altering system visibility. Highly scalable through automated tools. Defenders can obfuscate filenames and directory structures to hinder attackers' access to sensitive data.

DECOY [66]: Deploys false assets to mislead attackers. Moderately scalable, requiring realistic decoys. A common case is setting up a fake database with fictitious customer records.

REDIRECTIONS [67]: Redirects adversaries to fake targets to protect real assets. Highly scalable with automation. Suspicious traffic can be rerouted to honeypots for monitoring.

PERTURBATIONS [68]: Introduces noise or distractions to obscure real data. Moderately scalable but requires ongoing management. For instance, generating synthetic system logs can make it harder for attackers to extract meaningful insights.

The five layers that make up our taxonomy in a generic way has been previously presented. In Table II, presents the components of the Tactic, Strategy and Technique layers. It can be seen that each tactic can be related to one or more strategies and, in turn, these strategies can be related to one or more techniques. The different deception mechanisms can be classified using this table and the above taxonomy definition. Moreover, such a Table II has been developed to concisely

TABLE II
RELATIONSHIP OF THE TACTICS, STRATEGIES AND TECHNIQUES LAYERS OF THE PROPOSED TAXONOMY

Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion	Credential Access	Discovery	Lateral Movement	Collection	C2C	Exfiltration	Impact	Prevention	Detection	Reaction	Forensic		Phishing	Injection	Spoofing	Obfuscation	Honey-X	Poisoning	MTD	Honey-X	Obfuscation	Decoy	Redirections	Perturbations
					✓							✓	✓	✓	✓		Masking	✓			✓			✓	✓	✓		✓	
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						✓								✓	✓		Dazzling				✓								✓
		✓	✓			✓	✓							✓	✓	✓	Mimicking			✓					✓		✓	✓	
	✓			✓	✓									✓	✓		Inventing					✓			✓				
✓								✓	✓	✓	✓	✓	✓	✓	✓		Decoying		✓			✓		✓	✓	✓	✓	✓	
✓							✓	✓		✓				✓	✓	✓	Bait	✓									✓	✓	✓
								✓	✓		✓			✓			✓	Concealment											
								✓		✓		✓	✓	✓			✓	Camouflage			✓				✓	✓			
✓	✓	✓		✓								✓		✓			✓	False Inf	✓	✓	✓			✓		✓	✓			✓
✓	✓	✓		✓				✓				✓		✓			✓	Lies		✓				✓			✓			✓
					✓									✓	✓	✓	✓	Displays		✓						✓				

The figure uses two colors to differentiate the approaches: red represents attack approaches, while blue indicates defense approaches. On the left side of the figure, all attack and defense tactics are displayed, while the right side shows all defense and attack techniques. In the central section, deception strategies are highlighted in black font, serving as the connecting element between the tactics and techniques.

represent the different options available within the taxonomy. How this table contrasts with the current state-of-the-art will be discussed in Section VI-C.

Overall, the taxonomy comprises five interrelated layers (Approach, Dimension, Tactic, Strategy, and Technique) that enable comprehensive characterization of any cyber deception mechanism. The Approach layer distinguishes between offensive and defensive actions; the Dimension layer identifies the application domain (data, network, system, or software); the Tactic layer situates techniques within a specific phase of the attack or defense cycle; the Strategy layer defines the underlying objective of deception (e.g., concealment or decoy); and the Technique layer specifies the employed methods (such as phishing, honeypots, or disruptions). These layers operate in concert: the chosen strategy directs the appropriate tactic, which in turn determines the applicable techniques, all within the constraints of the selected approach and domain. With this comprehensive framework of components and relationships established, the next section examines how various proposals integrate these dimensions into complete defense architectures, assessing the distribution and synchronization of automation, monitoring, and response at each taxonomy level.

IV. CYDEC FRAMEWORKS

To address *RQ3* (refer to Fig. 2), it is necessary to determine whether a framework can use CYDEC in various dimensions, employing different strategies to fully defend specific assets using deception. To this end, we aim to answer research question *RQ3*. The search for articles based on deception frameworks or architectures has resulted in the articles shown in Table III, which are described below, grouped and introduced according to their similarities (Game Theory, SDN, etc.). In this table, the approach feature has been omitted due to the fact that no framework developing an attack approach has been found.

A. Current Frameworks

As a result of the state of the art research carried out in Section V-A, several frameworks dealing with CYDEC have been found. For this reason it has been concluded that an exhaustive study and comparison of these frameworks is needed. Below we detail each one of them.

In [69], a deception network (D-Net) methodology that redirects traffic from a compromised operational network (O-Net) to an identically configured deception network, minimizing further compromise of operational data and assets was proposed. Specifically, it uses packet rewriting techniques using SDN technologies, ensuring that the attacker does not notice the switch from O-Net to D-Net. This solution allows the attack to be observed and contained in a secure environment, continuing normal operations on the O-Net and studying the adversary's tactics without the adversary noticing.

In the article [70] introduced the Active Deception Framework (ADF), an extensible development environment designed to build sophisticated CYDEC applications. ADF utilized a SDN-based infrastructure that facilitated continuous monitoring and rapid response to adversarial activity. This framework provided a rich set of Application Programming Interfaces (APIs) that enabled the secure implementation, orchestration, and deployment of deception techniques, sparing security administrators from having to deal with low-level technical details. ADF was noted for its ability to create honeypot networks, distort attackers' perceptions, and deplete their resources through spatio-temporal mutation techniques. On the other hand, [71] presented a specific framework for Internet of Things (IoT), using honeynets to protect IoT devices from cyberattacks, especially IoT-specific attacks. This framework incorporated an OAuth authentication mechanism that verified the authenticity of requests and redirected suspicious traffic to a Honeynet for analysis, preventing attacks and allowing to capture and study attacker behavior, providing a double layer of defense.

Similarly, the work in [72] also focused on protecting IoT networks, but with a focus on mitigating Distributed

TABLE III
COMPARISON OF DIFFERENT DECEPTION-BASED FRAMEWORKS FOUND IN THE LITERATURE BETWEEN 2019 AND 2024

Ref	Year	Dimension	Tactic	AI	Orchestration	Zero-touch	UCs
[69]	2019	Network Data	Prevention Reaction	*N/S*	✗	*N/S*	✗
[73]	2019	Network	Prevention	✗	✗	*N/S*	Internet of Battlefield Things (IoBT)
[76]	2019	Network System	Detection Reaction	✗	✗	*N/S*	✓
[77]	2019	Network	Prevention	✗	✗	*N/S*	IoT 5G
[71]	2019	Network	Prevention	✗	✗	*N/S*	IoT
[70]	2020	Network	Detection Reaction	✗	✗	*N/S*	✓
[75]	2020	System Software	Detection Forensic	✗	✗	*N/S*	✓
[78]	2020	Network	Reaction	✗	✗	*N/S*	ICS
[79]	2020	Data	Reaction	✗	✗	*N/S*	*N/S*
[72]	2021	Network Data	Prevention Reaction	✗	✗	*N/S*	IoT
[80]	2021	System Data	Reaction	✓	✗	*N/S*	*N/S*
[74]	2021	Network Data	Reaction	✗	✓	✓	*N/S*
[81]	2022	Network Software System	Reaction	✗	✗	*N/S*	Internet of Vehicles (IoV)
[82]	2022	Network	Detection Reaction	✓	✓	✓	*N/S*
[83]	2022	Network System Software	Detection Reaction	✓	✓	✓	Cloud
[54]	2022	Network Data Software System	Prevention Detection Reaction Forensic	✗	✗	✗	Enterprise Mobile ICS
[84]	2023	Network	Prevention	✗	✓	✓	*N/S*
[85]	2024	Data Network	Detection	✓	✗	✗	*N/S*
[86]	2024	System Network	Reaction	✗	✗	✗	Cyber Physical System (CPS)
[87]	2024	System Network	Detection Reaction	✓	✓	✗	ICS
Ours	2024	Network Data Software System	Prevention Detection Reaction Forensic	✓	✓	✓	Generic

N/S (Not Specified) by the authors, ✓ addressed, ✗ not addressed by the analyzed work

Denial of Service (DDoS) attacks. This framework used SDN technology to redirect traffic and continuously monitor the network, enabling a proactive defense that significantly reduced the effectiveness of DDoS attacks by implementing defense strategies based on dynamic orchestration of network resources. Besides, Game theory optimized cyber defense strategies were proposed in [73] and [74]. Specifically, the former proposed a Partially Observable Stochastic Game (POSG) model to capture the dynamic interaction between a network defender and an attacker. In this context, the defender introduced honeypots strategically to deceive the attacker. In contrast, the attacker developed strategies based on imperfect observations of the network state, allowing to simulate and analyze how the decisions of both players affected the network security in real-time. Likewise, the latter used the CHIMERA framework, based on Partially Observable Markov Decision Process (POMDP), to autonomously plan and orchestrate CYDEC in real-time, classifying the adversary's actions into TTPs and creating a deception environment that sought to deflect, distort, exhaust, and discover the attackers' intentions.

Some studies, such as [71] and [70], shared the use of honeypots and CYDEC techniques to protect networks and devices. This two articles focused on analyzing and redirecting attacks using honeypots. In addition, [75] introduced Citrus, a novel method to generating attack signatures incorporating host-based telemetry extracted from honeypot endpoints. In particular, Citrus used a module called Tangerine to analyze information from intelligence sources, normalize data and generate signatures of malicious behavior, enabling rapid defense against emerging attacks. Meanwhile, the work in [76] presented an efficient honeypot architecture designed to capture high-quality attack data, using three main modules, decoy, orchestrator, and captor, coordinated to provide sensitivity, countermeasures and stealth.

Furthermore, the Internet of Vehicles (IoV)-specific framework, HoneyCar, was described in [81], configuring vulnerabilities in honeypots to deceive attackers and gather valuable information about their strategies and techniques. Besides, a shadow controller-based framework for protecting the SDN control plane against reconnaissance attacks, was

presented in [77], using shadow controllers to respond to reconnaissance traffic and improve network availability by constantly changing the attack surface.

In [83], an optimal defensive deception framework for container-based cloud environments was presented using Deep Reinforcement Learning (DRL). This framework orchestrated and updated decoy placement strategies dynamically to maximize attacker confusion and minimize actual attack paths, integrating adaptability capabilities in dynamic cloud environments. Another interesting technique called Decepti-SCADA is in [78], which introduced a framework designed for Supervisory Control And Data Acquisition (SCADA) systems using decoys and malicious interaction detection. Decepti-SCADA employed Docker containers to create highly interactive decoys that replicated real operating systems, using a Web-based management system to deploy and monitor these decoys.

The paper in [84] presented DOLOS, an architecture that unifies Moving Target Defense (MTD) and Cyber Deception techniques. DOLOS integrates deception and randomization techniques directly into production systems, avoiding traditional disadvantages. In particular, it uses agents deployed on each computer in the system, managed by a central controller. This architecture creates a dynamic and confusing attack surface, forcing attackers to interact with numerous fake services before compromising the real ones.

Additionally, authors in [80] presented a system that used CYDEC techniques to protect against various types of malware, creating deception playbooks that mapped malware behaviors to specific deception strategies. DodgeTron, described in [79], implemented a standalone CYDEC system that dynamically analyzed malware to discover attack techniques and build deception playbooks, providing a proactive and effective defense against advanced threats such as APTs.

In [54], the MITRE Engage framework was introduced to provide a structured approach for planning and executing adversary interaction operations, focusing on understanding and manipulating attacker behaviors while minimizing risk to operational assets. Unlike other frameworks, MITRE Engage leverages its alignment with the MITRE ATT&CK matrix to provide a comprehensive view of adversary strategies, enabling the orchestration of deception techniques across prevention, detection, reaction, and forensic tactics. The framework's unique "Ten-Step Process" offers practical guidance for iterative improvement, distinguishing it from frameworks like ADF [70] or DOLOS [84], which prioritize specific deception techniques or environments. However, MITRE Engage has notable limitations: it does not explicitly address the integration of AI for optimizing deception strategies, it primarily focuses on three predefined use cases, and it lacks comprehensive guidance on automation approaches like orchestration or Zero-touch deployment.

The Security Orchestration, Automation, and Response (SOAR) System in [82] is another orchestration framework for cyber deception that analyzes malware behavior by observing WinAPI calls and mapping them to MITRE ATT&CK techniques. From these observations, SOAR designs deception techniques to confuse the detected malware. Concretely, it slant is reactive, requiring an agent to detect running malware, which limits its effectiveness against rapidly mutating threats.

A CGAN-based cyber deception framework for ICS was introduced in [87], leveraging Conditional Generative Adversarial Networks to dynamically generate defensive network topologies. The framework integrates SDN to implement network shuffling and deception strategies, prioritizing system availability while effectively mitigating reconnaissance attacks. Extensive simulations demonstrated its efficacy in enhancing security and reducing computational overhead.

The Advanced Cyber Deception Framework (ACDF) in [85] combined honeypots and Tabular Masked Transformer (TabMT) models to generate realistic network traffic and deceive attackers. This framework automated the creation of diverse deception artifacts, improving honeypot authenticity and intelligence gathering. The approach emphasized adaptability and scalability, enhancing the robustness of cyber deception strategies against sophisticated threats.

A game-theoretic framework for multi-domain cyber deception was proposed in [86], addressing coordinated defensive strategies across cyber and physical domains. This framework modeled multi-layer deception games to ensure consistent and effective deception. Applications in modern warfare highlighted its potential to strengthen security in cyber-physical systems by integrating deception techniques in interconnected domains.

Based on a thorough analysis of the frameworks proposed in the literature discussed in Table III, the lack of a generic framework adaptable to various environments has been identified. In addition, it was noted that existing frameworks focus on specific deception techniques. Specifically, it was also noted that these frameworks do not take advantage of the different dimensions of deception identified (i.e., Data, Network, System or Software), or the different tactics of defense (i.e., prevention, detection, reaction or forensic). Furthermore, only a few of the most recent frameworks consider the use of advanced methods of deception. It is also observed that none of the current frameworks develop an offensive or attack approach when reacting to attacks or threats being a possible challenge to address in the future.

When compared, the various frameworks offer unique strengths and cater to distinct operational needs. MITRE Engage excels in structured planning and alignment with adversary behavior models, making it particularly valuable for organizations seeking to leverage threat intelligence systematically. ADF, with its rich API and automation capabilities, is more suited to dynamic environments requiring agile responses and extensive orchestration. DOLOS, by contrast, shines in scenarios where integration into production systems and attack surface randomization are critical for maintaining operational continuity under adversarial pressure. Frameworks like D-Net and CHIMERA contribute by emphasizing real-time interaction and decision-making, while HoneyCar and Decepti-SCADA highlight domain-specific applications such as IoT and SCADA systems.

While no single framework comprehensively addresses all dimensions of CYDEC, each provides specific advantages depending on the context. The choice of framework should

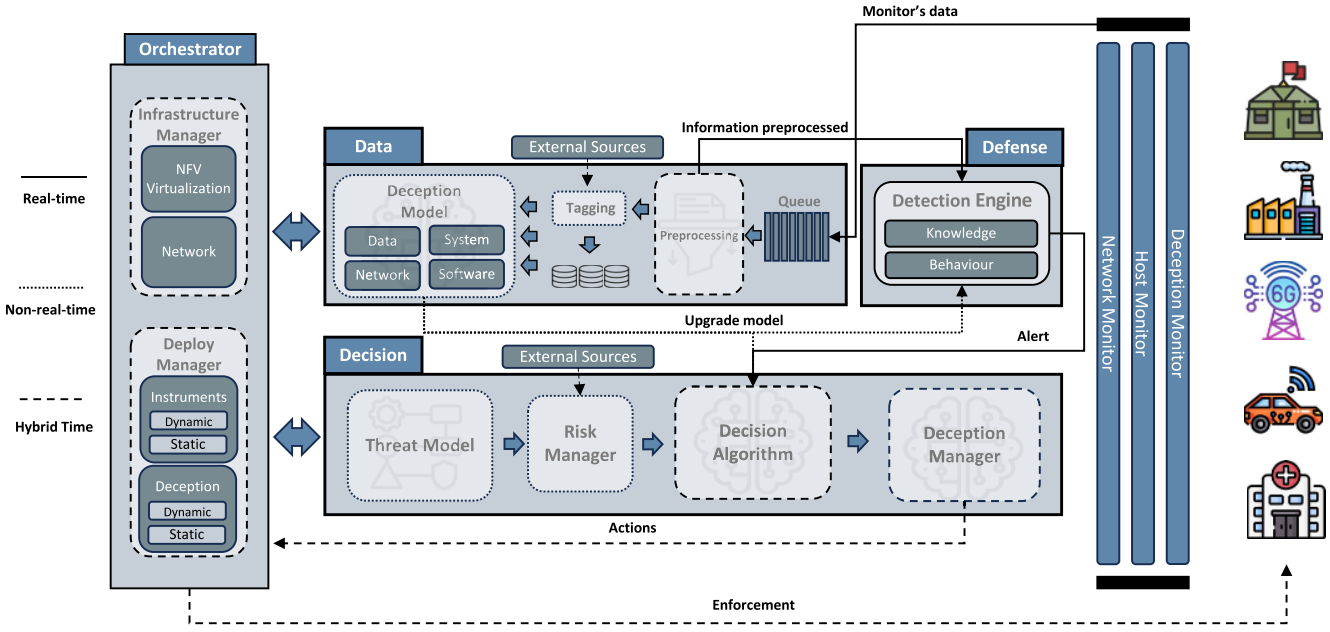


Fig. 4. Proposed framework for performing prevention, detection, response, and forensic tasks using CYDEC.

therefore be guided by the operational environment, the scale of deployment, and the desired level of automation and adaptability. Future work in this domain could focus on unifying the best aspects of these frameworks, integrating advanced AI-driven strategies and enhancing scalability to create a truly versatile and comprehensive deception framework.

Therefore, it has been decided to design a defensive generic framework that elevates deception to a higher level. This perspective will combine various techniques designed to prevent, detect, react, and facilitate forensic analysis, thus improving security mechanisms. In addition, advanced methods of AI will be incorporated to optimize and automate these techniques, ensuring a more robust and adaptive defense against emerging threats.

B. Our Proposed Framework

This section presents the proposed CYDEC based framework. Fig. 4 shows an abstract view of the resulting design framework, which consists of several layers capable of implementing and maintaining CYDEC-based security mechanisms. This framework leverages various CYDEC strategies and mechanisms in generic environments, making it versatile for applications in 5G networks, ICS, military settings, and beyond. Its adaptability lies in the inclusion of common components shared across different use cases, ensuring broad applicability and flexibility.

It should be noted that the proposed framework, unlike many others, uses AI mechanisms to enhance defense in all four tactics of defense (prevention, detection, reaction and forensics).

1) Layers: First, the layers contained in the framework, its main functions and components as well as tools that could be used to fulfill the functionality are described.

In Fig. 4, the names of the layers that compose the framework are represented as blue boxes, internally in a warm gray color the internal components that each layer has are represented.

DATA: This layer focuses on managing and manipulating data relevant to the system's security. Specifically, it includes defining deception models to protect the system against attacks, tagging data for identification and classification, storing and processing data in databases, and preparing data for further analysis.

- Queue: A data structure used to store and process security events in an orderly and efficient manner. Examples of queueing tools include Apache Kafka [88] and ZeroMQ [89].
- Preprocessing: This component consists of initial data processing before analysis to improve data quality or extract safety-relevant features. Some tools used for data preprocessing are TensorFlow [90] and Pandas [91].
- External Sources: External data is always necessary in any defense framework. The information we collect and obtain will be joined with external information from other data sources in order to create robust and efficient models.
- Tagging: It consists of assigning labels to data to identify its importance, sensitivity, or other characteristics relevant to the system's security. Tools commonly used for tagging data as Apache Atlas [92].
- Database: A data store used for various security functions such as event logging, identity and access management, etc. Some commonly used databases for security purposes include MongoDB [93] and Splunk [94].
- Deception Model: It is in charge of creating the AI models that will be used to implement the deception techniques in the different dimensions, i.e., with the collected and preprocessed data it will perform model training to provide the system with deception-based defense models. Some commonly used model algorithms are ML, Neural Network (NN), Large Language Model (LLMs), Natural

Language Processing (NLP), among others [95]. Within this component, several models will be created based on the dimensions seen in our taxonomy of Section III-B.
  - Data: Deception model based on simulation or characterization of real data.
  - System: Based on simulating system behaviors or actions.
  - Software: Model based on behaviors or actions of specific software to attract the attacker's attention.
  - Network: A model that performs packet sending or simulated behavior over the network to deceive the attacker.

DEFENSE: The defense layer performs threat detection and alerts the Decision layer to react to the threat.

- Monitor: Continuously monitors system health and security. They obtain real-time information from systems and networks and pass information to the data layer for preprocessing and analysis to detect suspicious activity in the organization.
  - Deception Monitor: it monitors the deception techniques implemented to warn of a potential threat. Tools commonly used for monitoring deception include TrapX [96], Illusive Networks [97], and Attivo Networks [98].
  - Host Monitor: it monitors hosts for potential threats. Examples of host monitoring tools include OSSEC [99] and Tripwire [100].
  - Network Monitor: It monitors the state of the network to detect suspicious behavior and warns of possible threats. Examples of network monitoring tools include Snort [101] and Suricata [102].
- Detection Engine: Component responsible for detecting potential threats or intrusions into the system. The detection engine uses security analysis techniques to identify malicious or suspicious activity patterns and generate alerts for further analysis. The information to be analyzed will be provided by the data layer so that it can be preprocessed for easier and more efficient analysis.
  - Knowledge: The knowledge base the detection engine uses to identify malicious or suspicious activity patterns. Some commonly used tools for knowledge-based detection include Splunk Enterprise Security, IBM QRadar [103] and Snort.
  - Behaviour: System and user behavior analysis to detect anomalies indicating a potential attack in progress. Examples of behaviour analysis tools as Darktrace [104].

DECISION: The decision layer is responsible for analyzing and evaluating security events to determine the appropriate actions to take, i.e., it is responsible for the direct protection of the system. These functions may include identifying and prioritizing potential threats, assessing the risk associated with these threats, and making automated decisions or recommendations to mitigate the identified risks. In addition, the decisions made will be based on CYDEC, i.e., it will choose which techniques or techniques to use on an asset at a given point in time.

- Threat Model: It describes the potential threats and vulnerabilities that could affect the system. The threat model identifies the system's information assets, the potential threats that could affect these assets, and the vulnerabilities that attackers could exploit. Some tools used for threat modeling include Microsoft Threat Modeling Tool [105] and OWASP Threat Dragon [106].
- External Sources: Used to obtain a correct risk assessment of each of the threats that may attack our system.
- Risk Manager: It assesses the level of risk associated with the identified threats and helps to prioritize security measures. This assessment will be based on various objective and subjective factors of the system. It will be used by the Decision Algorithm to correctly choose which hazards to mitigate and in what way. Examples of risk assessment tools include RiskLens [107] and FAIR-U [108], among others.
- Decision Algorithm: Algorithm used to make automatic decisions in response to security events. The decision will result in a series of steps performed by the Deception Manager to deploy the deception techniques. TensorFlow [90], and Scikit-learn [109] are commonly used decision algorithms.
- Deception Manager: Component used to specify the deception technique(s) to mitigate or protect a system. Once the decision has been made and the deception has been managed, this information will be sent to the Orchestrator that will be in charge of deploying everything. Tools commonly used for managing deception techniques include TrapX [96] and Attivo Networks [98].

ORCHESTRATION: The orchestrator layer coordinates and manages the different operations and resources of the infrastructure. This involves managing the infrastructure required to operate the system and deploy and configuring the different defense tools and techniques. The orchestrator ensures that all parts of the system work together efficiently to protect the system against cyber threats. Additionally, it employs Zero-touch automation, which minimizes or eliminates the need for manual intervention in routine processes.

- Infrastructure Manager: It manages and controls the infrastructure resources necessary for the system's operation.
  - Network Function Virtualization (NFV) / Virtualization: Responsible for carrying out the management of virtualization of defense mechanisms or system components, starting containers, virtual machines, etc. Some tools commonly used for this purpose include OpenStack [110] and Docker [111].
  - Network: This component is in charge of configuring and ensuring the proper functioning of the networks. Examples of tools used for network configuration include Cisco ACI [112] and OpenDaylight [113].
- Deploy Manager: It coordinates the deployment of the system's different defense tools and techniques. In other words, it is in charge of launching the different mechanisms to prevent, detect, and react to threats.
  - Instruments: It is responsible for deploying the artifacts necessary to ensure the system's security. These artifacts can be dynamic artifacts initialized in real time or static artifacts initialized in previous

states. Among the most commonly used tools are Snort [101], Suricata [102].
  - Deception: It main function is to deploy strategies designed to deceive potential attackers and deflect or deter their attempts to compromise system security. In the same way as the previous component, it can contain dynamic strategies in real time or static strategies in previous states. Examples of deception tools include TrapX [96], Illusive Networks [97], and Attivo Networks [98].

2) Flows: Different flows have been identified depending on whether the information and data flowing through the proposed framework are in real-time or not. The definition of these flows is intended to provide a way to defend reactively and proactively against potential cyber threats.

- Real-time: This flow is characterized by processing the data entering the system and performing actions in real-time without the need for manual intervention, i.e., Zero-touch. First, data is obtained from the monitoring layer based on Deception, Host, and Network Monitors and placed in a queue where it is preprocessed and evaluated in an orderly manner by the detection model located in the Defense layer. Whenever the detection engine detects suspicious behavior, an alert is issued to notify the decision model. The latter will decide on the optimal reaction strategy based on the information gathered in the alert, the metrics analyzed by the threat model, and the potential risk. This decision is then communicated to the Deception manager, which is responsible for implementing the strategy chosen by the decision algorithm. Finally, the orchestrator executes the actions specified by the Deception manager in the corresponding environment.
- Non-real-time: This workflow, which does not operate in real-time, consists of preprocessing the data, tagging it correctly, and storing it in the relevant database. Subsequently, AI models are created periodically to improve detection, decision, and deception systems. In addition, models capable of simulating realistic behavior are generated, which the Deception manager uses to implement preventive and forensic deception measures. These models are applied according to the required prevention need. The decision algorithm uses these models according to the environment's needs, and the Deception manager selects which model to use and where to implement it. Finally, the orchestrator executes the necessary actions in the corresponding environment.
- Hybrid Time: This workflow operates in both modes mentioned above. The information received can be in real-time or not, and the system will work in one or the other mode according to the needs and actions specified. For example, the actions taken by the Deception Manager and the Enforcement carried out by the Orchestrator can be a reactive action in real time or a preventive action in non-real-time.

The revised frameworks share the objective of articulating cyber deception techniques within cohesive architectures; however, none currently covers all application domains (data, network, system, software) nor fully embodies the four defensive tactics (prevention, detection, reaction, forensics) with a unified level of automation and adaptability. To address these gaps, a generic framework has been developed that integrates all dimensions and tactics, provides modular deployment, enables zero-touch orchestration, and incorporates AI to optimize defense at every stage of the cyber deception lifecycle. The following section presents a state-of-the-art analysis of works implementing various cyber deception methods, assessing their strengths, limitations, and technical maturity.

V. Survey Design

Based on the line of argument presented in Fig. 2, this section responds to questions *RQ4* and *RQ5* (refer to Fig. 2), where the main characteristics of the deception mechanisms present in the literature are defined and analyzed, as well as the AI methods most commonly used in CYDEC.

First, the methodology used to choose the articles to be analyzed is defined in Section V-A. Next, articles that do not use AI are examined. In these cases, the analysis focuses on the deception techniques they employ. These articles often resort to traditional methods, such as manipulation of information or the use of psychological techniques to influence readers' perceptions.

The key in this category is to understand how these articles are designed to deceive or manipulate without the intervention of advanced technologies such as AI. On the other hand, there are articles that do employ AI. These items are analyzed based on the type of AI they use. The main characteristic and innovation of these articles lies precisely in the use of AI. Unlike articles that do not use AI, the focus here is on understanding how AI technology is applied to achieve the article's objectives, which can include anything from automatic content generation to personalizing messages to influence the audience in a more sophisticated way. To conclude, current trends in the field of CYDEC are discussed based on the previous analysis.

A. Methodology

This section outlines the criteria for selecting the articles to be analyzed in the survey and describes the methodology used for their review. The methodology and the replicable process are illustrated in Fig. 5, which also shows the number of articles involved in each tactic. To carry out this process, two different criteria were used to exclude or include articles, i.e., objective and subjective selection.

In objective selection, the criteria are based on measurable and verifiable standards, ensuring transparency and consistency. This selection process is further divided into inclusion and exclusion criteria. For inclusion, articles containing "Cyber Deception" or "Deception" along with cybersecurity-specific keywords (cybersecurity, defense, attack, honeypot, reactive, MTD, etc.) in their title, abstract or keywords, published between 2019 and 2024 because there is currently no survey that collects all the articles on CYDEC and AI mechanisms in recent years. This interval lapse of time was

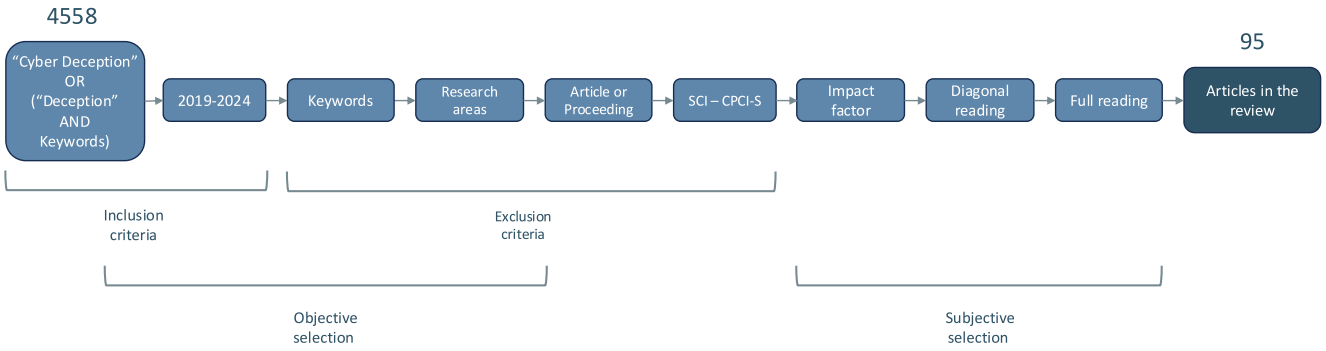


Fig. 5. Methodology used for survey paper selection.

chosen due to the recent increase in related articles, and the absence of comprehensive studies focused on CYDEC. The aim is to cover the main representative articles in this field, comprehensively addressing various CYDEC techniques and strategies. In addition, priority was given to articles indexed in reputable sources such as [114], [115] for their consistent quality.

Exclusion criteria were established to filter out articles that did not fit the research objective. To this end, articles based solely on keywords and those that his research area not related to engineering or computer science were excluded such as cognitive, psychological and other items. In addition, articles not published in journals or conferences were omitted, given their importance in academic research.

Subsequently, a subjective selection of articles was made based on personal perspectives. Therefore, journal articles with a medium or high impact factor (Impact Factor > 2) were considered valuable to the scientific community and retained. Subsequently, an iterative process of diagonal and close reading of the remaining articles was conducted to select those that focused on applying, designing, or evaluating CYDEC techniques or strategies.

After carrying out this methodology to select the papers, a total of 95 papers were identified. These papers are divided into three categories to analyze them coherently and correctly: CYDEC frameworks already analyzed, papers that design or implement some CYDEC mechanism without AI, and, finally, papers that design or implement some CYDEC mechanism using AI.

B. Features

Once the items to be analyzed are available, it is necessary to unify the analysis of each one of them to have a fair comparison. To this end, we defined the main characteristics that would allow for a systematic and comprehensive classification of the collected studies. These features were chosen through a twofold process: first, by examining commonly used features in previous works to establish a baseline for comparison; second, by identifying additional features critical to the domain of CYDEC based on the gaps observed in the literature and the requirements of a holistic analysis. This dual approach ensures that the selected features not only align with the established methods in the field but also address previously overlooked aspects, such as specific threats, maturity levels, and application scenarios. By including these elements, we provide a more detailed and nuanced understanding of CYDEC mechanisms. For instance, the taxonomy proposed in Section III serves as the foundation for classifying the deception mechanisms, while additional characteristics, such as Threats, Use Cases, and TRL, are introduced to expand the scope of the analysis. Below, we justify the inclusion of each feature, illustrating their relevance to both the classification and deeper understanding of CYDEC mechanisms.

1) Approach: Indicates whether the element focuses on defense strategies, as shown in Section III-A. Prior studies often overlook this distinction, grouping diverse mechanisms without clarifying their purpose. By incorporating this feature, we provide clarity on whether a mechanism is intended for proactive defense or as part of an offensive strategy [45], [46].

2) Dimensions: Dimensions refers to the level or layer of the computing environment at which security measures are applied or attacks are carried out, as discussed in Section III-B. As mentioned above, it can include the data, network, system, or software dimensions [47]. In particular, it helps to understand where security strategies are focused or where vulnerabilities or weaknesses are exploited. In this case, different dimensions can be chosen, as the same mechanism can be used in different environments. This feature was chosen to facilitate an understanding of the targeted vulnerabilities and the scope of applicability of each mechanism.

3) Tactic: This feature indicates the specific tactic of a technique or strategy associated, as discussed in Section III-C. Depending on the deception mechanism, it can include defense or attack approach. Concretely, it is chosen to identify in which tactic of the cybersecurity cycle the scientific work or study is focused. In the case of defense, the values that could be taken are prevention, detection, reaction, or forensics [52]. For attack, we rely on MITRE's ATT&CK tactics [51]. The same mechanism can be involved in several tactics of defense or attack depending on the deception mechanism used. Looking at the above surveys, none of them quantify which is the tactic where the different CYDEC mechanisms are used. While we identified a crucial aspect to describe the existing work.

4) Strategies: Strategies represent the specific strategies used in deception, as shown in Section III-D. Each CYDEC mechanism can be composed of several strategies. The value(s) that this feature may take are indicated in the taxonomy

proposed in Section III-D. In this context, strategies focus on identifying the purpose for which a deception mechanism is used, i.e., the goal it wants to accomplish [55], [56]. Depending on the objective that each author wants to give to the techniques used, the associated strategies will be different.

5) Techniques: This feature refer to the specific techniques used to carry out the deception, as discussed in Section III-E. In this case, several deception techniques may be used simultaneously to accomplish the same task or several. Depending on whether the article is based on defense or attack, the techniques will be one or the other. Different techniques can be used simultaneously to accomplish one or several objectives.

6) TRL: TRL indicates the maturity level of a specific technology, from initial concept to deployment and widespread availability. This characteristic helps to assess the feasibility and effectiveness of the technologies analyzed in the article. The main reason for using this feature is that we have identified that it is important to know the maturity level of the various deception mechanisms that currently exist to observe the current state of CYDEC. In addition, we wanted to see if it can be applied in real situations, because we think that CYDEC is one of the possible important weapons in the defense arsenal. This field's value is between 1 and 9 [116]. This feature was chosen because it provides an essential perspective on whether a mechanism is theoretical or ready for real-world deployment. By assessing TRL, we fill a gap in existing analyses, which often lack a focus on the maturity of technologies.

7) Use Cases (UCs): This characteristic indicates the context in which the CYDEC solution implemented or designed in the article will be used, i.e., the application scenario on which the deception mechanism is focused. Prior studies often generalize application areas, but our approach specifies the scenarios and provides examples of their implementation. This choice highlights the adaptability and relevance of CYDEC mechanisms to diverse environments.

- IoT: In the IoT scenario, deception mechanisms are tailored to protect interconnected devices and networks. Techniques such as IoT-specific honeypots simulate sensor data streams, creating realistic environments that mislead attackers. These techniques are significant as they address the unique vulnerabilities of low-powered, widely distributed devices, thereby offering a robust layer of security in environments where traditional security measures might fall short [117].
- Cloud: Cloud-based deception leverages virtualization and containerization to deploy decoys seamlessly across dynamic environments. By creating fake sensitive data and simulated cloud services, these mechanisms not only detect unauthorized access but also confuse adversaries attempting lateral movement. This approach illustrates the research focus on maintaining service continuity and protecting valuable data assets in modern cloud infrastructures [118].
- Military: Military applications require precision and adaptability, as deception mechanisms must contend with adversaries possessing significant resources. In this context, techniques such as simulating command systems and building fake communication networks are critical. These methods are engineered for high accuracy and robustness; they not only misdirect cyber adversaries but also provide valuable intelligence regarding the tactics and tools employed by threat actors [119].
- ICS: ICS have unique operational requirements. Here, CYDEC mechanisms simulate control networks and automation processes to shield critical infrastructure from disruption. In this sense, techniques such as deploying decoy SCADA networks and creating fake automation commands allow defenders to identify and isolate anomalies swiftly. These methods significantly impact the research context by demonstrating how simulation-based approaches can serve as both a defensive measure and an analytical tool [120].
- Enterprise: Enterprise security benefits from a multi-layered approach where application-specific honeypots, decoy databases, and simulated networks are deployed. The relevance of such techniques is reflected in their ability to integrate with existing security infrastructures, thereby enhancing overall network resilience without causing significant disruptions to regular operations [121].

8) Threat: Threat refers to the cyber threats or possible vulnerabilities addressed in the analyzed article, i.e., against which threats the implemented technique can defend [122] or the specific deception attack. Unlike previous works, our analysis explicitly maps threats to deception mechanisms, enabling a clearer assessment of their effectiveness. The range of values it can take is also very wide, as it could be used to mitigate any type of threat or attack depending on the deception mechanism used by the article. Some of the possible values are:

- DDoS: DDoS attacks aim to saturate network resources. CYDEC strategies mitigate these attacks by deploying fake servers and network segments that absorb or divert such an excessive traffic. The underlying technique often involves the dynamic reallocation of network resources to maintain service availability. Its significance lies in the dual benefit of detecting and neutralizing attack vectors while simultaneously gathering data to improve future defensive postures [123].
- Malware: Malware threats, including viruses and ransomware, can be addressed by isolating malicious code within controlled decoy environments. Techniques such as deploying decoy files or creating sandboxed systems allow security teams to safely analyze the malware's behavior, preventing its spread in the actual operational network. This controlled environment is vital for forensic analysis and contributes to the broader research context by deepening our understanding of malware propagation techniques and develop optimal countermeasures [124].
- APT: APTs present sophisticated challenges. CYDEC mechanisms counter these threats by establishing fake environments that deliberately mirror real operational systems. This procedure allows defenders to monitor attackers' strategies and tactics in a controlled setting. The technique's effectiveness in monitoring adversarial

behavior is critical for preempting data breaches and for informing the development of advanced defense systems [125].

- Reconnaissance Attacks: Reconnaissance attacks usually serve as precursors to more elaborate exploitation attempts. By deploying decoy systems that provide intentionally misleading information, CYDEC solutions obstruct the early stages of an attack. Here, Techniques focus on delaying adversaries, thus buying valuable time for incident response teams. Such an effort not only serves as a deterrent but also enriches the research context by offering insights into adversary behavior during the surveillance phase [126].

9) Use of AI: This feature indicates whether the research item leverages any AI methods. The values can be the algorithms used or the type of AI used. Mainly, we will focus on ML [127], DL [128] and Reinforcement Learning (RL) [129]. These algorithms have been chosen because they are the most commonly used AI mechanisms for defense. In addition, unlike the other works we include RL and DL in this analysis.

- Machine Learning: Algorithms such as Decision Trees and SVMs model behavioral patterns to build realistic decoys and detect malicious interactions.
- Deep Learning: CNNs and RNNs generate adaptive honeypots and synthetic data sets that evolve with attacker tactics.
- Reinforcement Learning: Agents learn optimal deception policies, through feedback from attacker engagement, adjusting strategies in real time.

VI. Survey

Once our methodology has been presented together with the feature for comparison, the following is the analysis of the articles identified in the methodology separated by CYDEC-based articles in Section VI-A to respond the *RQ4* (refer to Fig. 2) and CYDEC and AI-based articles in Section VI-C to respond the *RQ5* (refer to Fig. 2). In addition, Section VI-B presents the analysis of the articles analyzed in Section VI-A, and Section VI-D presents the analysis of the articles analyzed in Section VI-C. The analysis of the articles was carried out by observing the techniques presented by each one and internally grouping them according to the similarities between them. For this purpose, a description of each of the techniques used in each specific subsection has been included.

A. Techniques for a Non-AI Based CYDEC

First, the analysis of the articles that do not use any AI method has been performed. The result of the analysis of the characteristics identified above can be seen in Table IV. In addition, an individual analysis of each article according to the technique used was then performed. For this purpose, each article has been described in the section associated with each individual technique, commenting on the key points associated with the specific technique.

1) ***Honey-X:*** In this section, we will proceed to analyze several articles that focus on the use of Honey-X techniques. These include honeypots, which are systems designed to attract attackers and study their behavior; honeynets, which are complete networks of interconnected honeypots to observe more complex attacks; honeyfiles, which are decoy files designed to be accessed by attackers, allowing them to be traced; honeytokens, which are fake data inserted into systems to detect unauthorized access; and finally, honeydata, which refers to any type of fake information created for the purpose of being used as a trap.

Regarding **Honeypots** there are several studies using this type of technique. In the SDN arena, the authors of [130] focused on detecting and blocking botnet activities by integrating honeypots into the infrastructure to identify compromised nodes and enable dynamic network reconfiguration that quickly isolated threats using a novel intelligent blocking mechanism. This mechanism tracked relationships between nodes, specifically identifying and neutralizing loader nodes to prevent propagation, which reduced infection rates by 25% and increased the adversaries' wasted time fourfold. Similarly, the work in [131] used honeypots and game theory to anticipate and counter attacker strategies. Here, honeypots acted as decoys that deflected and tracked malicious activities, dynamically adjusting defenses based on game-theoretic models.

In the context of SCADA and ICS, the authors of [132] employed specific honeypots to lure attackers and study their behavior. This strategy induced threat-hunting activities through the use of deception. In addition, they introduced the concept of "decoy farm," a network of interconnected decoys that captured attackers' TTPs. By integrating this approach with kill chain analysis, they significantly improved detection of unknown threats and protection of heterogeneous SCADA environments. However, the reliance on static configurations created challenges in adapting to highly dynamic attack vectors. In [133], honeypot devices that mimicked real devices were implemented to induce malware attacks by sending fake control commands to detect and confuse attackers. Both studies focused on the importance of customizing honeypots for industrial environments, although they differed in the deception and detection techniques employed.

The authors of [134] was distinguished by its emulation of IP camera interfaces to tempt attacks, implementing Real Time Streaming Protocol (RTSP) and SSH services to interact with attackers and log their commands. This approach was specific to surveillance devices, providing a controlled environment to study attack techniques targeting IP cameras. A more general viewpoint was found in [135], which presented an improved high-interaction honeypot model designed to capture more attack information. Its virtualized implementation optimized resources and facilitated recovery after an attack, standing out for its high interaction and data collection capabilities.

In [136], honeypots were employed within digital twin environments to enhance the security of Operational Technology (OT) systems. These honeypots replicated real-world system components with high fidelity, attracting attackers and enabling defenders to analyze their methods in a controlled environment. This approach provided valuable insights into adversarial behavior while protecting critical assets from direct compromise.

TABLE IV
CLASSIFICATION OF NON-AI BASED CYDEC TECHNIQUES BASED ON OUR PROPOSED TAXONOMY. LEGEND IN FIG. 6

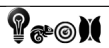
Ref	Year	Approach	Dimension	Tactic	Strategy	Technique	Threat	UCs	TRL
[168]	2019	Defense	Network	Reaction			Reconnaissance	Critical infrastructure	3
[138]	2019	Defense	Network	Prevention			*N/S*	IoBT	2
[147]	2019	Defense	Network	Detection Reaction			Botnets DDoS	*N/S*	3
[131]	2019	Defense	Network	Detection Reaction			*N/S*	*N/S*	1
[182]	2019	Attack	Data	Initial Access Execution Collection Defense Evasion			*N/S*	CPS	2
[183]	2019	Attack	System Data	Reconnaissance Initial Access Execution Defense Evasion			*N/S*	CPS	3
[118]	2019	Defense	Network	Reaction			APT	Cloud	3
[137]	2020	Defense	Network Data	Prevention			*N/S*	*N/S*	3
[184]	2020	Attack	Data	Initial Access Defense Evasion C&C Impact			*N/S*	ICS	4
[173]	2020	Defense	Network	Reaction			*N/S*	Wireless Network	2
[172]	2020	Defense	Network	Prevention Reaction			Reconnaissance	IoT	2
[165]	2020	Defense	Network	Detection Reaction			Botnets	*N/S*	1
[174]	2020	Defense	Network	Prevention			DDoS	Cloud IoT	3
[151]	2020	Defense	Software	Prevention			*N/S*	*N/S*	2
[169]	2020	Defense	Network Data	Reaction			APT DDoS	*N/S*	3
[152]	2020	Defense	Data System	Prevention Detection			Ransomware	*N/S*	3
[179]	2020	Defense	Data	Prevention			APT	*N/S*	3
[161]	2021	Defense	Network	Prevention			DDoS	IoMT IoT	3
[140]	2021	Defense	Data System Software	Prevention			Reconnaissance	Cloud	2
[171]	2021	Defense	Network Data	Reaction			DDoS Malware	Cloud	4
[176]	2021	Defense	Network Data	Prevention			APT	Military	3
[130]	2021	Defense	Network	Prevention Detection			Botnets	*N/S*	4
[132]	2021	Defense	Network	Prevention Detection			APT	SCADA	3
[159]	2021	Defense	Network	Reaction			APT	*N/S*	2

(Continued)

Another paper that addressed cyber defense from a hybrid angle was proposed in [137]. This study combined traditional security measures with CYDEC techniques such as honeypots, creating a layered defense against persistent attacks. Their approach leveraged both address mutation and fingerprint camouflage strategies to disrupt attacker reconnaissance and delay intrusion attempts. By dynamically coordinating these strategies, they significantly increased the time required for successful attacks while maintaining system reliability. Furthermore, their quantitative modeling of attacker behavior enabled fine-grained adaptation of defense mechanisms, further enhancing resilience against APTs.

Besides, in [138], it was discussed the use of honeypots in the context of game theory and prospect theory to design cyber defense strategies. It used these theoretical frameworks to model strategic interactions between a system and an attacker. Honeypots were employed as decoy devices to deceive attackers by directing them to unused nodes or false targets within

TABLE IV
Continued. CLASSIFICATION OF NON-AI BASED CYDEC TECHNIQUES BASED ON OUR PROPOSED TAXONOMY. LEGEND IN FIG. 6

[133]	2021	Defense	Network	Detection			Malware	ICS	3
[157]	2021	Defense	Software	Prevention			*N/S*	*N/S*	3
[162]	2021	Defense	Network	Detection Reaction			APT	Military	1
[175]	2021	Defense	Network	Detection			DDoS	*N/S*	3
[148]	2021	Defense	Network	Reaction			*N/S*	Power Grid Network	3
[181]	2022	Defense	Data Network	Prevention			Reconnaissance DDoS	*N/S*	3
[141]	2022	Defense	Network	Reaction			Reconnaissance	Military	2
[170]	2022	Defense	Network Data	Reaction Prevention			*N/S*	IoT IoMT	3
[139]	2022	Defense	Network	Prevention Detection			*N/S*	Cloud	2
[154]	2022	Defense	Data	Detection			Ransomware	*N/S*	3
[160]	2022	Defense	Network	Prevention Reaction			*N/S*	IoT	3
[166]	2022	Defense	Network	Prevention			APT	*N/S*	2
[158]	2022	Defense	Data	Prevention			Ransomware	*N/S*	2
[142]	2022	Defense	Network Software	Prevention Detection			*N/S*	IoT Military	2
[146]	2022	Defense	Data Software	Reaction			DoS	*N/S*	4
[155]	2022	Defense	Data	Detection			*N/S*	*N/S*	3
[145]	2022	Defense	Data Network	Forensic			Malware	Darknet	3
[144]	2023	Defense	Network System	Prevention Reaction			*N/S*	Cloud	2
[167]	2023	Defense	Network	Detection Prevention			APT DDoS	SCADA	2
[177]	2023	Defense	System	Prevention Detection Reaction			*N/S*	Military	3
[143]	2023	Defense	Network System	Prevention Detection			APT	IoBT	2
[180]	2023	Defense	Network Software	Prevention Detection			Reconnaissance	CMS	1
[135]	2023	Defense	Data	Detection			*N/S*	*N/S*	3
[150]	2023	Defense	Network	Detection Reaction			Reconnaissance	*N/S*	3
[156]	2023	Defense	Network Software	Prevention			*N/S*	*N/S*	4
[134]	2023	Defense	Network System	Detection			*N/S*	IoT	3
[185]	2023	Attack	Data	Execution			*N/S*	*N/S*	3
[153]	2023	Defense	Data	Prevention			*N/S*	IoBT	2
[163]	2024	Defense	Network System	Prevention Reaction			DDoS	IoT	4
[164]	2024	Defense	Data Software	Prevention Reaction			*N/S*	Critical infrastructure	3
[136]	2024	Defense	Network System	Reaction			APT	ICS	2
[178]	2024	Defense	Network System	Reaction			LDoS	*N/S*	4
[149]	2024	Defense	Network System	Prevention Detection			Reconnaissance	Critical infrastructure	4

a network. This was achieved through techniques such as deceptive routing and the installation of defense resources on specific nodes. Numerical results and simulations demonstrated the effectiveness of these strategies, optimizing the defense and complicating the attacker's decisions.

Additionally, game theory was further explored in [139], where possible attacks were modeled, and defenses were dynamically tuned using honeypots. This theoretical outlook enabled an optimized defense based on the anticipation of attackers' actions, highlighting the importance of informed and

adaptive defensive strategies. However, the effectiveness of the proposed Stackelberg game model relies heavily on accurate cost assessments, which may vary significantly in real-world scenarios. Thus, incorporating dynamic cost estimation frameworks could enhance the practical applicability of this model.

Continuous anonymization was presented in [140], which used CYDEC techniques such as honeypots, among others, to confuse attackers. This method focused on protection by hiding key information that attackers could use to identify and exploit vulnerabilities. Their proposed CACF algorithm dynamically adjusted container fingerprints in cloud environments, combining address hopping, fingerprint modification, and honeypot technology. This multi-faceted approach proved effective in thwarting reconnaissance attempts and increasing the cost of successful attacks.

Moreover, in [141], a hybrid system of low and high-interaction honeypots was introduced, using a hypergame model to deploy these honeypots effectively. This perspective focused on balancing the complexity of honeypots to deceive both novice and expert attackers, demonstrating the importance of a balanced defensive approach. Honeypot allocation and diversity were discussed in [142] and [143]. Both studies proposed using software diversification and strategic honeypot allocation to protect high-value resources in the network. These proposals highlighted the value of diversity and careful resource allocation in creating effective defenses.

The authors of [144] integrated agility and deception for proactive defense, using strategic honeypots and dynamic defense to complicate attack plans. Their MultiRHM methodology anonymized various attributes of network hosts, including IP and MAC addresses, as well as fingerprints, over temporal and spatial dimensions. This ensured the obfuscation of host identities, effectively disrupting reconnaissance and lateral movements during multi-staged intrusion attacks. Despite its robustness, the reliance on centralized mutation controllers posed challenges in scalability and security, particularly in highly distributed network environments.

Furthermore, in microservice architectures, the work in [118] used honeypots strategically to deceive attackers and collect valuable information about their tactics and techniques. By isolating containers and providing realistic interactions, the work ensured that honeypots were presented as legitimate targets, making it easier to capture detailed data and analyze attacks.

In addition, the work in [145] addressed the scalability and management issues of conventional IoT honeypots. It proposed a darknet-based proactive deception technique to improve IoT malware forensics by using a large network of honeypots to interact with infected IoT devices and collect malware artifacts.

In the paper [146], honeypots lured and deceived attackers by presenting themselves as vulnerable and valuable systems. When attackers interacted with these lures, their activities were monitored and recorded, providing crucial information about their methods and tools. In addition to capturing attack data, honeypots could launch controlled counterattacks, deflecting and misinforming attackers. This strategy enriched threat intelligence and enabled organizations to take proactive measures to strengthen their defenses.

The papers reviewed highlight the diversity and evolution in the use of honeypots as a central technique within cyber defense approaches. From hybrid models that integrate game theory to anticipate adversarial behavior to advanced anonymization methods such as MultiRHM, it is clear that a strategic and adaptive approach is key to counter increasingly sophisticated attacks. However, critical challenges arise in terms of scalability and operability. For example, although high-interaction based systems such as those proposed in [135] and [144] offer superior capability to capture detailed attack data, their deployment in distributed networks poses risks related to resource overloads and centralized points of failure.

Similarly, dynamic anonymization techniques, such as those explored in [140] and [145], show significant potential to confuse attackers, but their reliance on frequent configuration modifications may limit their adoption in high-traffic environments. These limitations underscore the need to develop integrated frameworks that combine the flexibility of decentralized systems with the robustness of current defense methods, ensuring not only effective attack detection and deterrence, but also performance optimization and ease of management in large-scale networks.

In connection with **Honeynets**, there are several studies that use this type of technique. In [147], a honeynet system composed of multiple honeypots specifically designed to attract and analyze DDoS attacks was presented. This attitude used IPTables to restrict outbound traffic and a Network Intrusion Prevention System (NIPS) to monitor and block malicious activity. The combination of these technologies allowed defenders to study DDoS attackers' tactics and develop better mitigation strategies, providing an additional layer of security through distraction and detailed analysis of attacks.

Additionally, the work in [148] implemented a network of dynamic honeypots, continuously configured and adjusted to make detection and prediction difficult for attackers. The constant reconfiguration of the honeypots ensured that attackers had difficulty establishing effective attack patterns, thereby increasing the power grid's security by complicating intrusion attempts.

In [149], the HoneyFactory architecture leveraged honeynets to create a multi-layered deception environment. The honeynets included diverse types of honeypots–traditional, simulation-based, and vulnerability honeypots–configured to simulate real business networks. This dynamic orchestration allowed for deep engagement with attackers, capturing complex attack patterns and enhancing the overall defensive posture of critical infrastructure.

The authors of [150] offered a comprehensive strategy that used honeynet to mitigate insider and external threats. This method displayed the implementation of Active Deception (AD) techniques to detect scanning and exploitation activities by attackers, redirecting their connections to a honeynet for analysis and protection of critical assets.

Moreover, in [151], honeynets were used to create deceptive and diverse networks that confused attackers. These networks included multiple software versions and configurations, luring attackers into controlled environments. By interacting with

these honeypots, their tactics were monitored and recorded, thus protecting real systems and providing valuable data to improve cyber defenses. Honeynets also restricted outbound traffic to prevent attackers from using compromised honeypots to launch additional attacks.

As for **Honeyfiles**, there are numerous studies that employ this methodology. In [152],the authors introduced PhantomFS, a file-based deception technology that creates fake files acting as decoys to lure malicious actors. These files are strategically placed and accessed through a hidden interface that prevents legitimate users and system services from mistakenly triggering alerts. The use of a dual-interface system (hidden and regular) enhances detection accuracy by isolating access to fake files exclusively through unauthorized activity, thereby reducing false positives. PhantomFS demonstrated minimal performance overhead while providing robust detection capabilities, making it a practical solution for environments prone to unauthorized intrusions.

In addition, the work in [153] extends the honeyfile paradigm to the IoBT. Honeyfiles are fictitious nodes that mimic real IoBT entities, generating false location data to mislead attackers and protect sensitive positional information. This method employs a combination of dummy IDs, fake packets, and a novel encryption scheme to obfuscate real locations and identities, validated through both mathematical modeling and NetLogo simulations. By enhancing the safety time and reducing the probability of attackers identifying real data, honeyfiles provide a scalable and efficient solution for securing dynamic and high-risk IoBT environments.

Similarly, in [154], distributed honeyfiles on the file system were employed to detect ransomware activity. This methodology relied on file access pattern analysis to determine the optimal locations of honeyfiles. When ransomware interacts with these fake files, the system can quickly identify the threat and trigger measures to mitigate the impact, thus protecting the real data from malicious encryption.

Also, the authors of [146] introduced trap documents as honeyfiles with malicious links designed to capture information from the attacker. In addition, Visual Basic for Applications (VBA) script files were used in Word and Excel documents, capable of executing reverse shells and evading security mechanisms. These honeyfiles detect attackers and enable counterattacks, obtaining valuable information about the machines and methods used by the attackers, which significantly enriches threat intelligence.

As regards **Honeytokens**, there are several researches that apply this type of technique. In [155], it was described its use, which included fake permissions, files, and documents, as part of an Attribute-Based Access Control (ABAC) system. These deception tokens were designed to attract malicious insiders by monitoring their activities on these fake objects to identify suspicious behavior. Integrating CYDEC into ABAC systems significantly improved the ability to detect insider threats and strengthened organizational security.

Moreover, the work in [151], honeytokens were strategically used to deceive attackers and detect malicious activity. These tokens, which could be fake credentials, or links, were distributed throughout the system. When an attacker interacted with a honeytoken, an alert was triggered, allowing defenders to track and analyze their actions. This method confused attackers into believing they had found valuable data and provided early threat detection, improving overall system security. In addition, the work [156] presented a novel technique to implement honeytokens in application traffic by means of a layer 2 network bridge. This allows it to function as a reverse proxy, but without the need for its own network address, facilitating its installation and integration without altering existing systems.

As for **Honeydata**, there have been multiple investigations using this technique. In [157], honeydata was used to deceive attackers by inserting fake data into different emulated software versions. This decoy data looked valuable and authentic, luring attackers and wasting their time and resources. By interacting with honeydata, attackers revealed their tactics, allowing defenders to detect threats quickly and protect real information. This point of view confused attackers and strengthened system security.

Besides, the work in [140], honeydata was used to deceive attackers in container environments. This fake data was strategically distributed within containers, capturing attackers to seemingly valuable but fictitious information. When attackers interacted with this honeydata, they revealed themselves to defenders, enabling early detection and rapid response. This technique not only confused attackers, causing them to waste time and resources but also protected real data by diverting malicious efforts to decoys.

In addition, the authors of [158] used game-theoretic modeling to anticipate ransomware strategies and protect critical data. A key technique in this technique was using honeydata to confuse attackers. This fake data acted as a decoy, diverting attackers' efforts and reducing the risk of real data being compromised or sold. This game-theoretic approach allowed for effectively anticipating and countering attackers' tactics, providing a robust defense against ransomware.

2) ***Decoy****:* In [132], researchers proposed an innovative strategy to protect SCADA networks using a "bait farm" composed of multiple fake systems simulating real SCADA devices. These baits lured attackers, diverting their efforts from genuine critical systems, and allowed defenders to observe and analyze their tactics in a controlled environment. By interacting with these decoys, attackers revealed their methods, significantly improving threat detection and response.

The works in [159] and [160] explored attack prevention through strategic baits. In the first paper, baits were designed to deceive attackers and divert their efforts, enabling defenders to train hidden Markov models to predict and detect malicious behavior. The second paper used fake IoT devices as bait to lure attackers and record their activities without compromising real systems. This interaction between attackers and defenders was modeled as a general sum game, optimizing the location and behavior of the baits to maximize threat detection and minimize risks.

Moreover, the authors of [161] used redirection as part of a proactive defense technique called MTD. It focuses on

protecting IoT networks by dynamically reconfiguring the network topology using decoys. Redirection is implemented by shuffling network connections at fixed or random intervals to divert attackers to fake nodes and increase the complexity and cost of attacks. This helps to prolong system lifetime and improve service availability by keeping attackers busy with decoys instead of real nodes.

Besides, the article [162] used decoys as a key technique within a deception-based defense platform for cyber threat detection in the context of hybrid warfare. The authors present an innovative model that integrates decoys at various levels of the network, including data networks, servers, and cloud environments, with the goal of deceiving attackers and gathering information about their tactics. These decoys are designed to resemble real assets, which increases their effectiveness by luring attackers and enabling early detection of their movements.

In [163], decoys were utilized within a MTD to protect IoT environments. The strategy focused on dynamically altering system configurations, including introducing fake nodes and misleading attackers with deceptive network components, thereby complicating their reconnaissance and exploitation attempts. Decoys not only lured attackers but also provided valuable intelligence on their tactics.

Similarly, the work in [164] proposed an optimized decoy selection scheme leveraging the MITRE ATT&CK framework. The decoys were strategically placed to intercept multiple attack paths with minimal resources, exposing the adversaries' techniques and creating opportunities to analyze their behavior while securing critical infrastructure.

Also, an innovative CYDEC technique called 2face, which used strategic disruption to combat botnets, was presented in [165]. Instead of taking down the botnet immediately, which would alert the attacker, 2face reverse engineers the botnet's Command and Control (C2C) protocol. With this information, realistic fake traffic is generated that fools both the C2C server and the bots into believing that the infection is still active.

Furthermore, the authors of [166] used baits as deceptive signals to divert attackers to false routes, mimicking the behavior of real systems. This allowed defenders to dynamically adjust their defense strategies and improve network security. Similarly, in [167] suggested implementing baits to protect Generic Object Oriented Substation Events (GOOSE) messages in digital substations, enabling the detection and mitigation of attacks without compromising critical substation communication.

Also, in the work [168] the use of decoys was described as a fundamental part of the dynamic deception system. This technique helps defenders save availability costs by keeping network services available to legitimate users while gathering critical information about attackers. The decoys are managed by a spoofing server that manipulates network traffic and simulates virtual resources to make the fake network appear authentic.

Moreover, the work [118] presented a system that confines suspicious microservices in a sandbox network to analyze ongoing attacks without compromising the production network. It uses decoys by cloning services in real time, maintaining a copy of the vulnerable production network. Sandnet implements three live cloning strategies: active, reactive and proactive. The deception technique is transparent to attackers, ensuring that they cannot detect the transition to the sandbox. The effectiveness of this strategy is measured by a metric called Quality of Deception (QoD), which evaluates the system's ability to keep attackers deceived about the real state of the network.

Also, in [156] described an innovative technique for inserting honeytokens into application traffic via a Layer 2 network bridge. These baits were invisibly injected into network traffic without needing their own network addresses, facilitating their integration and use without disrupting existing systems. Proof-of-concept implementations demonstrated how existing TCP packets could be modified and decoy HTML pages created.

Besides, the authors of [169] introduced "Mirage", a cyber deception system that uses decoys to replay in real time the network activities of a real system on a decoy server. Mirage is implemented as a TLS-capable reverse proxy that intercepts HTTPS requests from real users and duplicates them to generate requests to decoys. This approach allows maintaining an authentic access pattern on the decoys, making attackers believe they are interacting with a real server. To resolve inconsistencies between real servers and decoys, Mirage employs a decoy client emulator that handles stateful data features and caching logic. In addition, format-preserving encryption is used to obfuscate sensitive data before sending it to the decoy server.

In another study [142], the authors proposed a strategy for decoy allocation and software diversity implementation to improve network security. Baits were strategically placed to protect the most valuable resources, and a game-theoretic model was used to investigate the trade-off between software diversity and operational cost, demonstrating that careful honeypot allocation was critical to protect high-value nodes.

Moreover, the work in [170] combined differential privacy techniques with MTD and deception to protect statistical information in IoT networks. Baits helped to hide real network information, and a greedy algorithm was presented to balance defense cost and privacy protection level, proving effective in protecting the operational data of IoT networks.

Also, in [171], an active defense mechanism was presented that used CYDEC to redirect attacker traffic to quarantine machines and send spoofed responses in cloud-based data processing applications. Baits played a crucial role in real-time attack detection and the protection of sensitive data, proving to be a cost-effective, scalable, and easy-to-deploy solution.

Lastly, [172] employed CYDEC techniques and MTD to proactively change the network topology with real nodes and decoys. Baits helped prolong security failure and reduce the cost of defense, demonstrating their effectiveness in SDN-supported IoT networks under targeted attacks. The adaptive combination of these methods was shown to be superior to the fixed interval strategy in terms of decoy path ratio and mean time to security failure.

3) ***Redirection****:* In [118], the authors presented an innovative redirection technique that confined suspicious microservices in a sandbox network. Upon detecting suspicious activity, the system cloned the microservices and

redirected the attacker's traffic to these clones within an isolated network called Sandbox Network (SN). This viewpoint ensured that the attacker interacted with a copy of the real environment, allowing observation of their actions without compromising productive systems.

Conversely, the work in [161] utilized a redirection technique based on MTD. In this method, network configurations and accessible services changed dynamically, creating a constantly moving environment that confused the attacker. Redirection was achieved by implementing decoys and false targets, diverting the attacker's traffic to these deceptive elements, making it difficult to predict the network structure.

In addition, in [171] described an active defense mechanism called "defense by simulation" that uses redirection as a key strategy to protect cloud applications. This redirection is employed in the context of an attack, where detected malicious traffic is diverted to quarantine machines. By doing this, the system isolates the attacker's traffic, preventing it from reaching legitimate resources and causing potential damage.

Also, the authors of [162] took a more traditional slant by using redirection that was accomplished by directing attacker traffic to other components, allowing defenders to study attacker behavior in a controlled environment without compromising real systems. By integrating deception-based mechanisms, such as honeypots and redirection gateways, the system ensured that adversaries were engaged in a simulated environment, allowing the collection of valuable intelligence on their TTPs. This approach not only safeguarded real network assets but also enhanced the defenders' ability to understand and counteract evolving threats, demonstrating its relevance as a cornerstone of proactive cyber defense strategies.

Moreover, in [130] combined SDN capabilities with honeypots to redirect botnet traffic. This system leverages SDN's programmable control plane to identify suspicious traffic in real time and redirect it to honeypots for analysis. Furthermore, the intelligent blocking mechanism tracks the relationships among botnet members, enabling the system to target loader nodes specifically, which are responsible for botnet propagation. This targeted approach effectively reduces the infection rate by up to 25% and increases the adversary's wasted time by a factor of four.

In addition, in the work of [148] a new honeypot system based on deception defense technology was proposed to protect power grids. This system used a redirection technique where unused IP addresses were collected from the power grid to construct dynamic virtual hosts. When an attacker attempted to access these virtual hosts, the system was proactively responding or redirecting attack traffic to the honeypot in the background. This redirection deceived and trapped the attacker, extending the monitoring reach and improving defense against unknown attacks.

Furthermore, the authors of [169] proposed a technique to increase the credibility of decoys by recreating real network activities. In this case, redirection involved diverting malicious traffic to decoys that faithfully replicated the behavior of the original network. This tricked the attacker into believing they were interacting with real systems, while they were being observed in a secure environment.

In addition, the work in [173] presented the Deceptor-in-the-Middle (DitM) technique. Here, redirection was performed by intercepting and redirecting the attacker's traffic to deception Virtual Network Operators (VNOs) without the attacker knowing. This method allowed VNO operators to learn about attacks and the attacker's targets without compromising security or quality of service.

Besides, a hybrid mechanism combining redirection with early attack detection was proposed in [137]. In this case, redirection was achieved by creating fake access points that mimicked real ones, diverting suspicious traffic to these decoys, and allowing their analysis without risk to authentic systems.

The study in [147] proposed a honeynet-based defense mechanism to mitigate DDoS attacks by redirecting malicious traffic to a decoy network. The honeynet system was designed to absorb and analyze attack traffic while ensuring the protection of critical network resources. The implementation relied on advanced data control mechanisms, such as IPTables for restricting outbound traffic from compromised decoys and an intrusion prevention system to invalidate known attack signatures. Additionally, the honeynet facilitated extensive data capture, allowing for detailed analysis of attacker behavior, including packet inspection and the reconstruction of attack patterns.

Additionally, the authors of [138] used game and prospect theoretical models to develop redirection strategies based on manipulating attacker behavior. In this context, redirection was executed by creating scenarios that diverted the attacker's actions to false targets, using game-theoretic principles to anticipate and counter the attacker's moves.

Lastly, [168] proposed a dynamic redirection model based on the attacker's capabilities. This methodology adapted redirection in real-time according to the attacker's profile and abilities, diverting their efforts to lures designed to exploit specific weaknesses. This method allowed for a personalized and more effective response against different types of attackers.

4) ***MTD****:* In [161], several MTD techniques were explored for IoT environments. A prominent method was Shuffling-Based MTD, which involved changing network and system configurations like IP addresses, MAC addresses, and port numbers to confuse attackers and invalidate their reconnaissance data. The authors also used Diversity-Based MTD, which increased the attack complexity by employing various system components, such as different software or operating systems, offering the same functionalities. Lastly, Redundancy-Based MTD ensured system reliability by dynamically using multiple replicas of system components, guaranteeing operational continuity despite attacks.

In [137], MTD techniques were combined with CYDEC strategies to counter persistent scanning and foothold attacks. Address mutation was used to dynamically change IP addresses. At the same time, digital fingerprint camouflage techniques were applied so that decoy hosts appeared as legitimate systems. Coordinating these strategies dynamically

interrupted attackers' attempts to establish a foothold in the network. In addition, attackers' traffic was redirected to decoys, which wasted their resources and allowed the collection of intelligence on their methods.

Additionally, in [174] evaluated the effectiveness of combining Shuffle and Diversity techniques in cloud environments. Shuffle-Based MTD randomly changed the location of data and applications in the cloud infrastructure, while Diversity-Based MTD introduced variability in software and hardware components used. This combination made it difficult for attackers to predict asset locations and increased the cost of their attacks. The study showed that this combination could improve cloud system security and resilience without significantly affecting user performance.

Besides, the authors of [175], an innovative method was introduced involving the rotation of system components. This strategy continuously changed key system elements like the operating system and Web applications to make them more challenging for attackers to identify and exploit. Rotation was performed at regular intervals or in response to specific events, increasing attack complexity and reducing the attacker's success chances. Research demonstrated that this technique could be effectively implemented in SDN environments, providing an additional security layer.

Furthermore, the work in [176] proposed integrating MTD with SDN to create a dynamic cyber terrain. This technique used multiple MTD techniques, including IP address rotation, operating systems, and Web applications. Continuously altering the cyber terrain increased the time, resources, and skills required for an attacker to evade detection.

In addition, the application of MTD on drones was evaluated in [177] using game theory to optimize defense. Drones continuously changed their routes and communication patterns, making it difficult for attackers to predict their behavior. This perspective protected individual drones and ensured the integrity of the drone network. Specifically, game theory was used to model and analyze the interactions between attackers and defenders, helping identify optimal strategies for MTD implementation.

In [172] used SDN technology to proactively change the network topology using real and decoy nodes. This approach aimed to address the computational and resource constraints of IoT devices, which make conventional high-overhead security measures unsuitable. The mechanism incorporated MTD and CYDEC techniques to proactively counter threats by continuously altering attack surfaces, supported by a hierarchical graphical security model for analyzing the impact of single-target and multi-target attacks. Simulation results highlighted that adaptive shuffling of nodes outperformed fixed interval shuffling, achieving a higher proportion of decoy paths, significantly extending the mean time to security failure, and reducing overall defense costs.

Similarly, the work in [170] proposed a differential privacy-based defense method integrating MTD and deception techniques to protect statistical information in IoT networks. Using SDN to manage data flows and support shuffle-based MTD, the authors designed deployment strategies (random and smart) and a greedy algorithm to balance defense cost and privacy protection levels. That is, experimental results demonstrated the method's effectiveness, highlighting that the smart strategy was more cost-effective than the random one.

Also, the authors of [150] combined the capabilities of MTD and honeypots to improve enterprise network security. This system focuses on detecting and countering adversarial activities such as network scanning and service exploitation. By redirecting all attacker interactions to a honeynet, it safeguards critical infrastructure while collecting detailed intelligence on attacker behaviors. Notably, the use of proactive and reactive port-switching mechanisms adds a dynamic layer of deception, effectively disrupting the attacker's reconnaissance efforts and complicating their exploitation strategies. These port-hopping techniques not only obscure the network topology but also enable defenders to actively mislead adversaries, forcing them to expend additional resources and time.

The work in [178] proposed a novel defense strategy based on connection partitioning and MTD to mitigate Low-rate Denial of Service (LDoS) attacks. By employing containerization and live migration, the method iteratively isolates malicious connections by redistributing them across multiple server replicas and evaluating their health. Experimental simulations demonstrated its effectiveness, salvaging up to 80% of benign connections within 5 minutes, even without prior knowledge of the attack's specifics.

Moreover, in [179], a new dynamic data technique integrating deception-based MTD was proposed. In this sense, this technique changed the size of the data, its authenticity, and access privileges, significantly expanding the attack surface traversal space. The method provided dynamic access to data based on the attributes of users and their operations, protecting sensitive data and reducing system load without imposing significant costs.

Additionally, the work in [144] focused on defeating enterprise intrusion attacks through host identity anonymization using MTD. By dynamically changing host identity parameters, it increased the difficulty for attackers to map and compromise target systems. Defense was achieved through multidimensional, multiparametric host identity anonymization, which periodically changed host identities to confuse and delay attackers, improving overall system security.

Besides, in [166] proposed an MTD-enhanced cyber defense method based on signal games. The defense was improved by converting real IP addresses of hosts into virtual IP addresses and generating numerous decoy nodes to build a virtual network topology. This extended the attacker's network recognition time. The mechanism strategically alternates IP addresses among three policies: shuffle, shift, and hold, which ensure the continued survival of decoy nodes without disrupting the attackers' connections. Additionally, the framework incorporates signal game theory to model the attack and defense processes, allowing defenders to choose optimal deception strategies based on probabilistic revenue analysis. Despite its effectiveness in increasing attack complexity and reducing real network exposure, challenges such as computational overhead and scalability in large networks remain areas for further improvement.

Finally, in [160] proposed an IoT organizational deception model based on an adaptive general-sum competitive game, employing MTD and decoys to select appropriate periods,

intensities, and mutation targets in the IoT domain. The MTD strategy perturbed the attacker's cognitive judgment and built erroneous defense intelligence, using sampling techniques, shuffling schemes, and reactive tactics to optimize mutation in the IoT domain and perform deep deception based on systems.

5) ***Obfuscation****:* In [180], obfuscation techniques were implemented to confuse vulnerability scanners. A key method was "version trucology", presenting fake versions of plug-ins, themes and the Content Management System (CMS). HTTP status codes were also altered to hide components or make them appear different. For example, a component could respond with a 404 code even though it was operational. In addition, fake entries were added to the *robots.txt* file to divert scanners to irrelevant paths.

Moreover, the work in [169], "Mirage", a system that replicates network traffic in real time to create realistic decoys, was developed. Mirage operates as a TLS-capable reverse proxy that intercepts HTTPS requests from real users and duplicates them to decoy servers, maintaining seamless interactions and preserving state consistency. To address the challenges of inconsistent states and potential user data leakage, the system integrates a decoy client emulator and employs format-preserving encryption (FPE) to obfuscate sensitive information before replaying it to decoy servers. This ensures that decoys mimic real servers not only in appearance but also in behavior, thwarting evasion attempts based on network traffic or system fingerprinting. Mirage's ability to replicate authentic user interactions and maintain stateful HTTP features makes decoys indistinguishable from real servers, thereby increasing the complexity for attackers attempting reconnaissance or lateral movement.

Also, the study in [170] explored the protection of IoT networks against data leakage using obfuscation techniques. They protect statistical information in VLANs by combining differential privacy with proactive defense mechanisms such as MTD and deception. Using SDN technology, data flows between devices are managed and mix-based MTD is supported. Two deployment strategies were implemented: random and intelligent, theoretically testing differential privacy adherence to protect operational information of IoT networks.

Furthermore, in [181], obfuscation techniques were implemented to protect Web servers against fingerprinting attacks. Server responses were modified to appear different by altering HTTP headers and other indicators. Noise was also inserted into network traffic patterns to confuse attackers. These strategies made it difficult to obtain reliable information about the system.

Besides, the work in [153], obfuscation was explored to protect location privacy in IoBT by deliberately altering real location information with false or modified data. Also, in [146], an obfuscation technique was introduced that generated active decoys that interacted with attackers. These decoys collected information and generated fake responses and behaviors. Fake files and data were created to divert attackers' attention and generated fictitious network traffic and apparent system changes.

Finally, in [144] introduced the MultiRHM approach, an advanced obfuscation technique that employs a dynamic mesh of random honeypots combined with host identity anonymization across multiple layers. By continuously randomizing IP addresses, MAC addresses, and host fingerprints, this method disrupts attackers' reconnaissance and mapping efforts, preventing the reuse of gathered intelligence during multi-staged intrusions. The architecture integrates centralized mutation controllers and distributed gateways to manage real-time randomizations seamlessly, ensuring minimal disruption to legitimate network traffic while effectively redirecting adversarial interactions to decoys.

6) ***Perturbation****:* The work in [175] presented an innovative perturbation-based defense approach designed for DCmGwords was presented, focusing on the power converter controllers of these networks (critical, programmable elements that allow dynamic modification of their control gains). This strategy, called CMTD, involves proactively altering the gains at fixed intervals to prevent attackers from predicting and exploiting system behavior, thereby introducing variability that makes false data injection and replay attacks much more difficult as adversaries' models rapidly become obsolete.

Moreover, the authors of [140] addressed the perturbation technique by using continuous anonymization of container fingerprints in a cloud environment. The authors proposed a method called Continuously Anonymizing Containers' Fingerprints (CACF), which continuously modified container fingerprints to maintain an anonymization standard. Specifically, this technique involved dynamically changing IP addresses, ports, and other container attributes to prevent attackers from identifying and profiling critical services. The goal was to make the fingerprints of containers indistinguishable, creating an environment where each container appeared to be another, significantly increasing the difficulty for attackers attempting effective system reconnaissance.

Additionally, in [177], a perturbation technique applied to drone-based defense systems was presented. The authors used game theory to model defense, where drones continuously changed routes and communication patterns. This constant perturbation prevented attackers from predicting drone behavior and launching successful attacks. By frequently changing trajectories and communication channels, drones created a dynamic and highly unpredictable environment, increasing the cost and complexity of attacks targeting these systems.

B. Analysis

Once all the selected articles have been analyzed using the features, an exhaustive discussion of the set of trends observed is carried out in Table IV with the help of Fig. 6. The first thing that stands out is the great scarcity of articles that develop CYDEC techniques with an attack approach. One of the reasons could be that it is more common to do research in the defense field than in the offense field and, in addition, the articles that are based on offense focus on the ICS or CPS.

Besides, most CYDEC technologies are in intermediate stages of development, namely TRL 2 and TRL 3. These stages indicate that techniques are being conceptualized and experimentally tested. The prominence of these levels suggests a rapidly developing field in which many new techniques

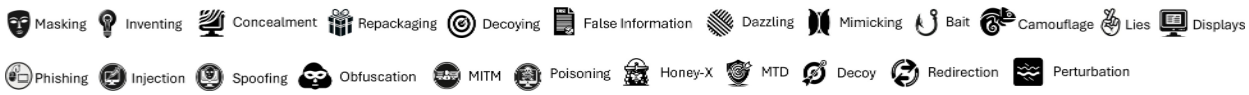


Fig. 6. Legend to Table IV representing the strategies and techniques of CYDEC.

are being validated and refined. The smaller presence of technologies at TRL 1 indicates that there are relatively few technologies in the early stage of research and development. In contrast, TRL 4, although less frequent, indicates that some techniques are progressing toward practical application in controlled environments.

Furthermore, the use cases for these technologies are varied, ranging from critical infrastructure to military environments to IoT devices. This diversity reflects the wide range of areas requiring robust security measures, as discussed in Section II-B. Furthermore, CYDEC is well suited to many environments, so it was decided to create a generic framework presented in Section IV. Critical infrastructure applications, such as energy and water, underscore the importance of protecting vital systems against attacks that could have a major social and economic impact.

Cloud security also features prominently due to the increasing use of cloud services for data storage and processing. Likewise, the use of CYDEC in military environments and systems highlights the need for advanced security in defense and industrial control applications. The mention of wireless networks and the prevalence of *N/S* (Unspecified) entries suggest that the research covers both specific and more general applications, where the specific details of the use cases are not always specified.

In addition, the threats that CYDEC technologies can mitigate are varied and include both common and advanced cyber-attacks. Among the most common threats are DDoS attacks, which seek to overwhelm a system with traffic to make it inaccessible, and APT threats, sustained attacks typically carried out by well-funded and organized actors, such as nation states. Other major threats that CYDEC can mitigate include malware, designed to damage or exploit systems, and ransomware, which encrypts victims' data and demands a ransom for its release. CYDEC is especially effective against reconnaissance attacks, which seek to gain information about a system before launching a more significant attack, as it can deploy decoys and fake systems that confuse and mislead attackers. The prevalence of *N/S* (Unspecified) entries suggests that there are still many areas where greater specificity in threat identification is needed, which may reflect both the breadth of research and the continuing evolution of cyber threats.

For the sake of the reader, Fig. 7 shows the relationship between the tactics in which the different CYDEC mechanisms are used and the dimensions on which they act. On the left side of the figure, a pie chart with this distribution is presented. On the right side, a bar chart shows the percentage of utilization of the different dimensions within each tactic. The x-axis of the bar chart focuses on representing the number of total items that are part of each dimension. The y-axis presents the existing distribution of dimensions by each tactic of the defense. In addition, the percentages within each of the dimensions represent the distribution that exists within each of the tactics of the defense.

Last but not least, there is a strong focus on the design and implementation of CYDEC mechanisms that rely on the network in the three main tactics of defense (i.e., prevention, detection and reaction). This trend indicates that threats to the network infrastructure are considered high risk and require more attention. A balance is also observed in the Data dimension, especially in prevention and forensics, reflecting the need to protect sensitive information and the ability to investigate compromised data. Consequently, the software and systems dimensions are not as prominent as the network and data dimensions. This tendency could suggest areas for improvement in security strategies. In addition, there is a serious lack of CYDEC generation mechanisms that rely on forensic data to feed systems and continuously improve them.

In addition, we note that there is no work that addresses the totality of strategies and techniques to defend against threats. This leaves an open door for future work that could implement or design combined strategies to create a comprehensive CYDEC mechanism.

In Fig. 8, all the CYDEC strategies and techniques present in the analyzed articles and their distribution over the CYDEC defensive techniques that have been analyzed are represented. In particular, a great predominance of the Honey-X technique can be observed, as it is well established in the literature and has been studied for many years as well as matured over time. The other techniques are present in a very varied form such as MTD, Decoy or Redirection, indicating that there is a diversity of approaches used for the different strategies.

Looking at the right side of the figure, we can see how the different techniques are distributed within the strategies, with a great predominance of the Honey-X technique. Finally, it is observed that, unlike the Honey-X technique, the techniques more associated with the data part, such as Perturbations and Obfuscation, are less explored due to the predominance of the network dimension that is observed in Fig. 7.

C. AI-Based CYDEC

First, we provide a brief primer on AI in cyber deception, introducing its key concepts and common techniques. Then, following the research questions presented in Section 2, CYDEC articles using an AI method have been analyzed in Table V. In particular, this section is organized by the branch of AI employed in each of the analyzed articles.

AI IN CYBER DECEPTION: A PRIMER AI has transformed cyber deception by endowing systems with the ability to learn from vast telemetry streams, adapt to adversary behavior, and optimize deceptive countermeasures over time. In the simplest form, traditional machine

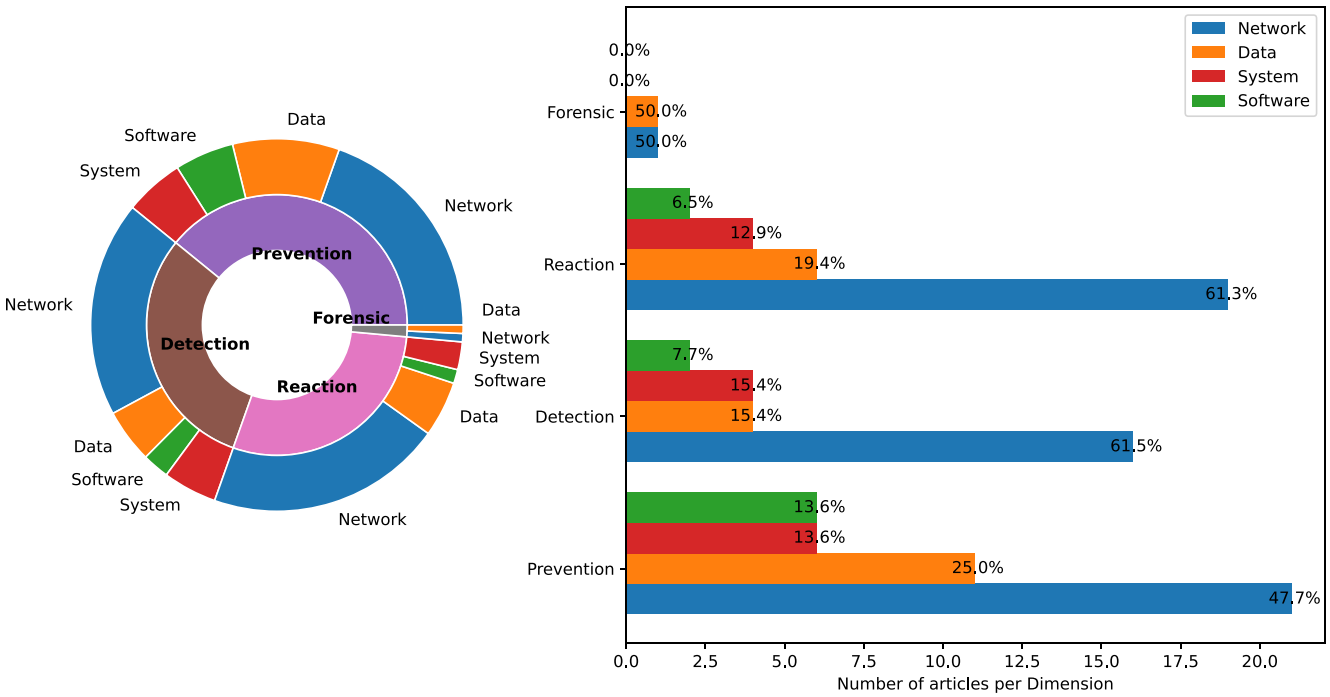


Fig. 7. Distribution of CYDEC mechanisms by the tactic in which they are used and their present dimensions.

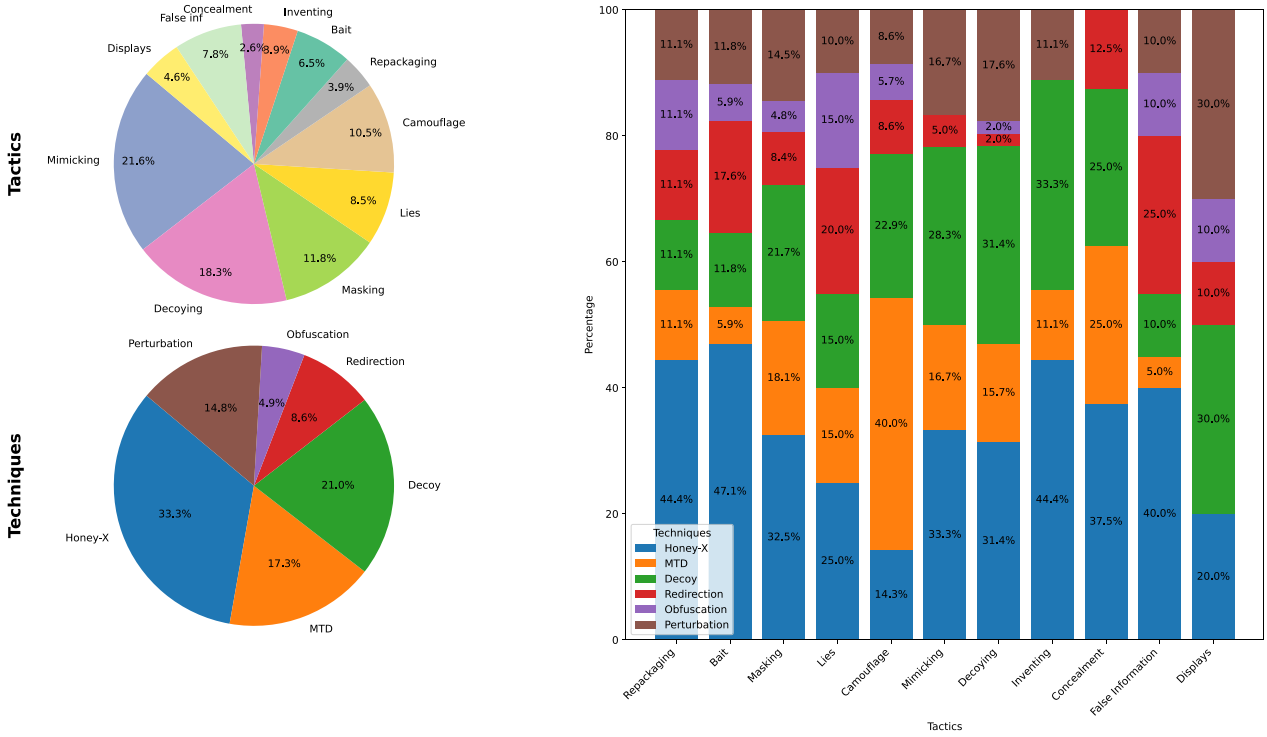


Fig. 8. Distribution of CYDEC mechanisms according to their strategies and techniques.

learning algorithms analyze labeled and unlabeled network or host data to detect anomalies indicative of attacker activity, automatically triggering or steering threats toward honeypots.

DL extends this capability by leveraging multilayer neural architectures (such as recurrent networks for temporal patterns or autoencoders for compact representations) to generate realistic decoy artifacts and synthesize phishing lures that evolve alongside attacker tactics.

RL makes deception a closed-loop strategy: agents interact with real or simulated environments, get feedback on attacker engagement and resource costs, and learn exactly when and where to deploy or retract traps to maximize engagement while minimizing overhead. Meanwhile, evolutionary and swarm

TABLE V
CLASSIFICATION OF CYDEC MECHANISMS THAT USE AI BASED ON OUR PROPOSED TAXONOMY

Ref	Year	Dimension	Tactic	Strategy	Technique	AI	Threat	UCs	TRL
[192]	2019	Network System	Detection Prevention			DL	DDoS	CPS	3
[191]	2019	Data Software	Prevention			DL	*N/S*	*N/S*	3
[193]	2019	Network	Detection			DL	*N/S*	*N/S*	2
[198]	2020	Data	Reaction			RL	*N/S*	*N/S*	1
[189]	2021	Data	Prevention			ML	APT	*N/S*	3
[187]	2021	Data	Detection			ML	*N/S*	Military	3
[195]	2021	Network	Prevention			DL	*DoS*	Wireless Network	3
[197]	2021	Network	Detection			RL	APT	ICS	3
[188]	2021	Data	Detection			ML	*N/S*	*N/S*	3
[196]	2022	Network Data Software	Detection			DL	*N/S*	Web IoT	3
[199]	2022	Network	Prevention Reaction			RL	*N/S*	*N/S*	3
[186]	2023	Network Data Software	Detection Reaction			ML	APT	*N/S*	2
[190]	2023	Network	Prevention Detection Reaction			ML	APT	IoT	2
[202]	2023	Network	Prevention			RL	*N/S*	*N/S*	2
[201]	2024	Network	Reaction			RL	*N/S*	*N/S*	4
[194]	2024	Network	Prevention Detection			DL	Reconnaissance	UAV	4
[200]	2024	Network System	Reaction			RL	APT	Cloud	4

N/S (Not Specified) by the authors

algorithms evolve distributed decoy configurations; transfer and few-shot learning adapt deception models to new network contexts with minimal data; and self-supervised contrastive methods learn normal behavior without labels to detect subtle deviations.

NLP methods power advanced social engineering decoys (chatbots masquerading as insiders or dynamically crafted document trails) while knowledge graphs and symbolic reasoning automate the creation of coherent, multi-stage bait scenarios by linking threat indicators and maintaining logical consistency across deceptive narratives. Altogether, these AI-driven approaches convert static defenses into intelligent, adaptive ecosystems that anticipate adversarial objectives, tailor responses in real time, and continually refine their strategies as attackers evolve.

1) Machine Learning: ML techniques are widely used in CYDEC for their ability to detect, mitigate, and deceive attacks through structured, data-driven approaches. A recurring theme among ML applications is anomaly-based detection. For example, [186] achieved a remarkable 99% accuracy and recall in identifying anomalous sessions in RDP event logs using AdaBoost, making it a tool for detecting and neutralizing APTs. Furthermore, the integration of ML in dynamic deception models exemplifies its role in enhancing efficiency. By employing belief update algorithms and tools such as Hidden Markov Models (HMMs) for policy allocation, defenders dynamically adapt to evolving threats. The use of hybrid encryption and socket synchronization ensures secure environments, illustrating the synergy between ML and real-time defensive strategies.

Similarly, DTs in [187] identified implausible positional data in space environments with 93% accuracy, revealing hidden satellites. This approach unveiled hidden satellites masquerading as debris, emphasizing the practicality of ML in uncovering deceptive behaviors even in highly complex contexts such as Space Situational Awareness (SSA). Both this and the aforementioned studies underline the profound capability of ML in identifying irregularities with remarkable precision, though they highlight a critical dependency on high-quality labeled data for optimal performance. Both studies underline ML's capability to identify irregularities with precision, though they highlight the dependence on high-quality labeled data.

Deception techniques also play a central role in ML-based CYDEC. In [188], the ProTO system demonstrated an innovative approach to combating adversarial topology inference. By manipulating polling packets, it reduced attackers' accuracy in network topology mapping by 99%, effectively deceiving attackers and disrupting their operations without significantly degrading network performance. Complementing this, [189] introduced EDGE, a system employing NLP to generate highly convincing deceptive content. This technique not only

masked sensitive information but also misdirected attackers by providing believable yet fabricated data, reducing the success rate of targeted attacks by 51.55%. While ProTO focuses on obfuscating network-level operations, EDGE excels at the content level, illustrating the versatility of deception in enhancing cybersecurity through resource misdirection and operational disruption. Both approaches underscore the adaptability and effectiveness of integrating deception with ML to fortify cyber defenses.

Beyond detection and deception, ML enhances strategic defense. Reference [190] combined hyper-theoretic game theory with ML to optimize multi-attacker defense strategies, demonstrating reduced attack success rates. Overall, ML's contributions to CYDEC showcase its versatility in identifying anomalies, crafting deception, and strategizing defenses. However, the dependency on data quality, scalability, and adaptability to novel threats remains a common challenge.

2) Deep Learning: The work in [191] presented a framework for generating deceptive data using DL and differential privacy. Private autoencoders created "HoneyData" indistinguishable from real data, making it difficult for attackers to identify fake data. This approach enhances deception by utilizing private autoencoders to learn latent structures, ensuring attackers cannot easily differentiate between real and fabricated data. The integration of differential privacy further strengthens the framework by protecting sensitive information while maintaining high data utility. For example, HoneyData generated through this method can be strategically deployed in compromised environments, forcing attackers to analyze misleading information and thereby reducing their operational effectiveness.

Besides, the authors of [192] used RNNs to model and simulate physical processes, improving deception fidelity in control system environments. This technique improves deception fidelity by embedding realistic, physics-driven behavior into cyber-physical decoys. For instance, in a simulated boiler system, RNNs were used to predict temperature and pressure variables, enabling the creation of decoys that behave indistinguishably from real system components. These high-fidelity decoys not only delay attackers by requiring them to expend more resources in reconnaissance but also increase the likelihood of detecting their presence through monitoring their interactions with these decoys.

In addition, the work in [193] proposed a decentralized framework using high-interaction honeypots embedded with decoy neural networks. These honeypots actively attract adversaries, capturing their tactics while protecting the primary system. The use of adversarial honeypots demonstrates how AI can create realistic and appealing targets, effectively diverting attackers and gathering intelligence for future defensive strategies.

Furthermore, the work in [194] introduced an unsupervised Graph Neural Network (GNN) approach to counter UAV network reconnaissance attacks. This method uses a GNN model to identify optimal fake UAV network configurations, misleading attackers into interacting with decoy networks while preserving the integrity of real network configurations. The adaptability of GNNs allows for the dynamic adjustment of deception strategies based on evolving threat landscapes, illustrating how AI can enhance both the efficiency and precision of defensive measures.

Finally, [195] investigated the use of DL for CYDEC in wireless networks. Concretely, it regulated the allocation of power to users and communication channels, tricking attackers into attacking specific channels and ensuring security on others. Experiments showed a 50% reduction in the effectiveness of DoS attacks, improving the security of wireless communications.

DL expands on ML's capabilities by processing complex, high-dimensional data, enabling advanced deception and detection strategies in CYDEC. A key strength of DL lies in its ability to generate highly realistic decoys. For instance, [192] used RNNs to simulate physical processes, creating decoys that blend seamlessly with real system operations, while [191] employed private autoencoders to generate 'HoneyData" indistinguishable from genuine data, leveraging differential privacy to enhance security.

DL also excels in dynamic attack detection and response. Reference [196] achieved a 99% accuracy rate in classifying HTTP requests, redirecting malicious traffic to deception environments orchestrated via Docker containers. In contrast, [193] utilized decentralized honeypots to attract and analyze attackers' attempts to manipulate DL models. These methods demonstrate DL's adaptability, whether through centralized traffic redirection or distributed engagement frameworks. In wireless communication security, DL methods like those in [195] and [194] further highlight its effectiveness. By regulating power allocation across channels and leveraging GNNs for deceptive configurations, these studies misled attackers and reduced the success of attacks, emphasizing DL's role in predictive modeling and dynamic adaptation.

3) Reinforcement Learning: RL stands out in CYDEC for its dynamic adaptability, allowing systems to learn and refine defensive strategies in real-time. A prominent application of RL is in adaptive honeypots.

For example, [197] introduced Asguard, an RL-based adaptive honeypot that dynamically adjusts its behavior to interact with attackers effectively while safeguarding against system compromise. By leveraging Q-learning, Asguard achieves a balance between engaging attackers to collect data and protecting the system from exploitation, recovering over 97% of attacker commands. This real-time adaptability enhances the effectiveness of deception by responding to attackers' evolving tactics, showcasing how AI can make honeypots more resilient and intelligent. RL also benefits from its integration with game theory.

In [198] developed a differentially private game-theoretic approach using RL to obfuscate system configurations and deploy deception assets. This method strategically misleads attackers by creating uncertainty in network structures, reducing their utility gains by 30-40% while preserving defenders' privacy. The integration of RL with game theory not only enhances the precision of defensive maneuvers but also ensures these strategies are adaptable to both static and dynamic network conditions.

In wireless networks, RL approaches like [199] optimized energy allocation using UAVs and Intelligent Reflective Surfaces (IRS), misleading attackers into targeting specific channels. This strategy reduced attack success rates by 40%, highlighting RL's ability to enhance resource management while incorporating deception tactics. Despite its adaptability, RL shares some challenges with DL, including computational intensity and the need for realistic training environments. Nevertheless, its real-time learning capabilities provide a distinct edge in scenarios where attackers continuously evolve their strategies.

Additionally, [200] presented a differential game-based deception asset selection algorithm leveraging DRL. By modeling attacker-defender interactions as differential equations, the study demonstrated real-time optimization of deception asset placement in a cloud-native environment. This approach models attacker-defender interactions using differential equations to dynamically allocate deception assets, significantly enhancing the speed and precision of decision-making. By applying DRL, the algorithm demonstrates how AI can adaptively manage complex, real-time threat scenarios, illustrating practical advancements in deception techniques.

Finally, [201] proposed a Nash Q-learning-based framework for deception asset deployment in cybersecurity. This multi-agent stochastic game model emphasized dynamic attacker-defender interactions in lateral movement scenarios, achieving over 70% defense success rates and showcasing the adaptability of Nash Q-learning in real-time deceptive defense strategy optimization. The adaptability of Nash Q-learning in real-time decision-making exemplifies AI's potential to revolutionize the strategic deployment of cyber defenses.

D. Analysis

After analyzing all CYDEC articles using AI in Table V, as can be seen in Fig. 9, it is identified that in the realm of CYDEC, during the threat detection tactic, the techniques of RL and DL are predominantly utilized. This trend reflects their effectiveness in identifying anomalous behavior and potential threats within deceptive environments. The ability of these techniques to process and analyze large volumes of data in real-time makes them ideal for detecting malicious activity on networks and systems camouflaged by deception strategies. Additionally, RL is employed in some detection cases, suggesting its applicability in environments requiring continuous adaptation and enhancement of detection accuracy within the deceptive frameworks.

In the prevention tactic of CYDEC, ML and DL are equally prevalent. These techniques are leveraged to anticipate incidents and prevent attacks before they occur by using the deceptive data to mislead attackers. The predictive capabilities of ML and DL, based on the analysis of historical patterns and trends within deceptive contexts, enable security systems to proactively anticipate and mitigate potential threats. In addition, the reaction tactic is highlighted by the use of RL, which is crucial for learning and adapting quickly to incidents in real-time. RL's ability to make adaptive decisions and continuously improve in response to new threats, even those obfuscated by deception strategies, makes it a valuable tool in incident management and post-attack recovery.

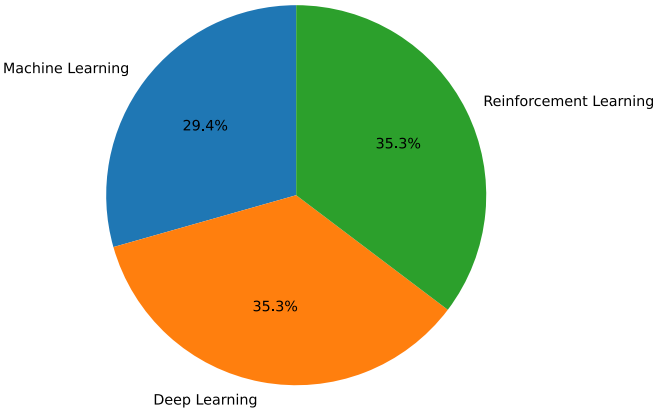


Fig. 9. Distribution of the different AI mechanisms used in CYDEC.

Fig. 9 shows the generic distribution of techniques present in the analyzed articles, where it can be seen how RL and DL techniques predominate.

It is clear that the use of AI in CYDEC is currently quite low. This is because it is just beginning to explore and take advantage of the enormous potential that this technology can provide. In this context, the use of AI is mainly focused on specific techniques such as Honey-X, reflecting a trend similar to the use of CYDEC without AI. This situation highlights the lack of research and development oriented to attack techniques, which is an area of opportunity for future studies and applications.

Furthermore, it is notable that in the defensive domain, solutions based on reaction and forensic analysis are considerably less prevalent than those focused on prevention and detection. However, it is important to note that responsiveness and forensics are critical components to a comprehensive cybersecurity strategy, and their lower presence may indicate areas where more research and development is needed to balance the cyber defense approach.

Additionally, ML and DL are prominent at the network layer. The ability of these techniques to process large volumes of data and detect complex patterns is crucial for maintaining network security. RL is also used at this layer, especially for real-time management and response, adapting to evolving threats.

At the data layer, ML is commonly used for pattern analysis and anomaly detection. Protecting sensitive data and identifying unauthorized access are key areas where ML demonstrates its effectiveness. At the systems and software layer, although AI usage is lower compared to network and data, ML and DL are still present, analyzing vulnerabilities and strange behaviors in systems and applications.

Most AI technologies in CYDEC are at TRL 2 and TRL 3 levels, indicating that they are in advanced stages of testing and development. This direction suggests a rapidly developing field where techniques are experimentally validated before practical adoption. RL also appears at these levels, indicating experimental implementation and validation in controlled environments.

Moreover, Use cases for ML and DL include applications in IoT and military environments, reflecting the need for data and pattern analysis in large data volumes and critical contexts. Security in IoT is crucial due to the proliferation of connected devices, while in military contexts, AI must protect sensitive and operational information.

Specific threats such as DDoS and APT attacks are commonly mitigated with DL and ML. These techniques are effective in detecting and mitigating massive and sophisticated attacks, respectively. RL and ML are used to identify and respond to reconnaissance activities, adapting quickly to counter these information gathering attempts. Many studies do not specify particular threats, but the common use of ML and DL suggests their broad and flexible applicability in a variety of security situations.

The integration of AI into CYDEC significantly enhances the efficacy and precision of deception techniques due to its ability to analyze complex patterns, predict attacker behavior, and adapt to evolving threats in real-time. Traditional deception strategies often rely on static configurations or pre-defined rules, which can become predictable and less effective against sophisticated attackers. In contrast, AI enables dynamic and context-aware deception by continuously learning from data and refining tactics to exploit attacker tendencies [203].

VII. Lessons Learned

After reviewing and analyzing the state of the art, the following lessons have been learned:

Imbalance between Prevention, Detection, Reaction and Forensic The analysis shows a predominant focus on prevention and detection, leaving a significant gap in reaction and forensic analysis capabilities. This can be attributed to the desire to prevent incidents from the outset, but highlights a missed opportunity in terms of post-attack learning. Improving reactive capabilities with dynamic and adaptive analytics could significantly strengthen cyber defenses. To close this gap, embed end-to-end forensics at deployment: auto-capture packets and system calls, feed them into a replay engine, and generate root-cause reports so defenders can instantly trace and patch exploited flaws.

Effectiveness of CYDEC Techniques The analysis of CYDEC techniques reveals clear trends in their effectiveness across different defense tactics. Honey-X stands out as an integral and versatile technique, excelling in both detection and prevention, making it critical for highly sensitive environments like enterprise networks. Decoy-based techniques perform well in prevention and reaction, showing their value in real-time attack mitigation scenarios. On the other hand, Redirection and MTD focus on reaction and prevention, respectively, emphasizing their adaptability in disrupting attacker persistence. Techniques like Perturbations and Obfuscation, while scoring lower overall, act as complementary measures suitable for delaying attackers in specific contexts. Lastly, the forensic application of Honey-X, although less frequent, demonstrates its unique ability to capture valuable attacker information, underscoring its multifaceted utility. Define clear SLAs (e.g., 99.9% uptime and sub-second Honey-X activation, 10-minute perturbation rotations) to continuously measure and tune each technique's performance against evolving threats.

Lack of Offensive Techniques in Research Studies focused on offensive techniques are scarce. This is probably due to the ethical and legal sensitivity of such research. However, understanding how attackers exploit vulnerabilities is crucial to building more effective defenses. Controlled development of offensive approaches, such as simulations in closed environments, could provide valuable insight. Incorporate deception into red-team sprints: track metrics like time-to-first-touch and dwell time on decoys, then refine lure fidelity until adversaries consistently engage for a target duration.

Predominance of Network Dimensions CYDEC techniques show a strong focus on the network dimension, reflecting the perception that network infrastructures are prime targets for attackers. However, this has led to an underexploration of the systems and software dimensions, equally critical areas that could benefit from more research attention. Extend deception to endpoints by deploying lightweight agents that report execution traces, and embed hidden feature-flags in applications that only reveal decoy content under malicious probes.

AI integration in CYDEC The incorporation of AI methods, such as ML, DL, RL and other represents an important opportunity. However, their adoption is still in early stages, with applications limited mainly to specific techniques. As these technologies mature, their integration is expected to enable more dynamic and adaptive solutions, capable of anticipating and effectively responding to complex threats.Treat AI models as evolving assets: shadow-deploy new classifiers, monitor prediction drift, and promote only those policies that demonstrate superior engagement and low false-lure rates.

Difficulties in the Generalization of Solutions A recurring challenge is the lack of solutions that are both specific and generalizable. Current proposals tend to be designed for particular contexts, such as IoT or military environments, but lack adaptability for different domains. The development of more generic mechanisms could close this gap and enable broader applications. Provide a deception SDK with declarative APIs for bait generation and traffic steering, so new scenarios in cloud, IoT, or on-premise environments require only simple configuration changes.

Impact of Technology Maturity Levels With most technologies at TRL 2 and 3, the field is in an experimental phase. While this reflects rapid growth, it also highlights the need to validate and bring these solutions into real-world environments. Accelerating this transition will be key to maximizing CYDEC's impact in practice. Run staged pilots under "chaos" tests (e.g., fault injections and traffic spikes) to validate decoy resilience, availability, and alert latency before scaling to production.

Predominance of use cases Once the articles have been analyzed, it can be seen that most of them do not specify the use case where the techniques are applied. However, among those that do, their application in Cloud, IoT and military environments predominates, which highlights the

dual capability of CYDEC (military and civil). This trend also evidences the lack of generalization of the techniques for implementation in multiple environments simultaneously. Establish a quarterly review of documented scenarios versus real incidents, retiring obsolete decoys and fast-tracking new lures for emerging attacker techniques.

TECHNIQUE-SPECIFIC LESSONS LEARNED With Honey-X, we have found that simulating realistic environments allows us to detect changes in attacker tactics without compromising our critical resources. With MTD, the constant variation in configurations and directions has forced adversaries to readjust their strategies, increasing the cost and complexity of their attacks. Also, Decoy has shown us that the use of decoys transforms intrusion attempts into information sources, allowing us to extract key data on the methods employed. The Redirection technique has made it possible to channel malicious traffic to controlled areas, facilitating the study of attack behavior without affecting the operability of the system. With Obfuscation, intentionally modifying responses and attributes has generated uncertainty for the attacker, making it difficult to identify critical components. Finally, Perturbation has revealed that constant small changes in essential parameters hinder attacker recognition and planning.

VIII. OPEN CHALLENGES

To address *RQ6* (refer to Fig. 2) and based on the current state of the art, we have found several weaknesses and challenges that could be addressed to improve deception-based defenses. The following points represent the main challenges that future CYDEC solutions might consider.

ADVANCED AI INTEGRATION IN CYDEC The use of AI in CYDEC is surprisingly underrepresented in the literature. Less than half of the reviewed articles use AI, indicating a lack of its potential in this field. AI integration can significantly improve deception mechanisms by enabling continuous adaptation and constant improvement. AI-based systems can learn from previous attacks and adjust deception strategies in real-time, making mechanisms more resilient and effective.

Currently, new methods and forms of AI have emerged, such as federated architectures and generative models, which are capable of improving results in various contexts. However, in the analysis, no article has been found that uses these advanced technologies in the CYDEC domain. This weaknesses represents a significant challenge but also a unique opportunity to drastically improve our AI mechanisms in this field. In this sense, federated architectures enable collaborative learning without centralizing data, increasing privacy and security [204]. This methodology can be extremely valuable in CYDEC, as it allows the sharing of information about threats and attack strategies without compromising sensitive data. To implement this methodology, federated networks could be established where each participant maintains control of its own data, using techniques such as federated learning and secure information sharing. In addition, generative models can enhance existing CYDEC methods through their creative capabilities [205]. To achieve this goal, generative response models could be created to improve the efficiency of the responses currently proposed in the literature.

In addition, AI can facilitate real-time analysis of the attack, enabling accurate attribution of activities to different cyber criminals. It is possible to identify specific behavioral patterns and relate them to known actors using advanced data analytics and ML techniques. Research should focus on incorporating federated learning and generative models to enable decentralized, privacy-preserving collaboration.

COORDINATED INTEGRATION OF CYDEC TECHNIQUES In the current context of how authors carry out deception, there is a tendency to use solutions that implement deception techniques individually or in pairs [206]. Several CYDEC techniques have several internal variations well documented in the literature. However, rarely are all of these techniques employed together to launch in parallel against attackers or defenders.

This lack of integration and coordination among multiple CYDEC techniques represents a major opportunity in the field. The simultaneous application of multiple techniques and their variants could exponentially improve deception outcomes. Employing multiple strategies in a coordinated manner increases complexity for the adversary, making detection and evasion of deception more difficult.

The joint use of techniques can also greatly benefit decision-making. New algorithms and systems developed in the current literature make it possible to process large amounts of data and make decisions in real-time. Integrating various CYDEC techniques can significantly improve the ability to track and analyze the attack and threat. This, in turn, allows selecting the most appropriate deception method in each specific situation.

In addition, the parallel implementation of multiple techniques can provide broader and deeper coverage of the threat environment. For example, while one technique may distract the attacker, another may gather valuable information about his/her strategies and targets. Combining different methods creates a more robust and effective deception network, increasing the likelihood of confusing and manipulating the adversary. Developing unified platforms that harmonize multiple deception tactics is fundamental. Establishing interoperability standards and creating decision-making algorithms can support the coordinated deployment of these techniques, enhancing overall defense effectiveness.

THE IMPORTANCE OF STEALTH IN CYDEC From our perspective, an effective CYDEC mechanism must possess stealth as its main characteristic, as the essence of deception lies in its invisibility. A solution that cannot operate unobtrusively, whether in an attack or a defense scenario, loses its effectiveness. If such a solution is detected, the surprise factor vanishes and, with it, the ability to influence the adversary's behavior.

In exploring the current literature on CYDEC, we find a surprising absence of the word "stealth" in the proposed solutions. In our opinion, two main reasons can explain this omission. On the one hand, the solutions developed so far may fail to remain stealthy, suggesting a significant technical limitation. Without the ability to operate in the shadows, these tools cannot effectively deal with real threats that are

increasingly sophisticated and capable of detecting intrusions and manipulations.

In addition, likely, researchers have not characterized stealth as a crucial component in their solutions. This point of view could be due to a lack of emphasis on the importance of stealth within the CYDEC field or an underestimation of detection's impact on the effectiveness of deception. Ultimately, stealth must be the pillar on which CYDEC solutions are built and evaluated. Ensuring that deception systems remain unobtrusive is critical.

FORENSIC TACTIC IN CYDEC Another point little explored by the literature analyzed in this survey is the forensic tactic. In this tactic, CYDEC is used to generate sufficient intelligence to improve our systems and protect us in a better way for each attack received and mitigated. The intelligence generated from these forensic analyses becomes a powerful tool for continuously improving security systems. Each attack analyzed provides lessons learned that can be applied to strengthen defenses against the methods used in the recent attack and against variations and possible future threats.

However, the current literature does not address this critical tactic sufficiently, likely because it often focuses more on immediate responses and technical details rather than on the strategic application of insights gained from past incidents to prevent future attacks.

Leveraging forensic analysis to extract actionable intelligence from deceptive interactions will improve future defenses. Automated forensic tools that feed data into a continuous improvement loop can help cybersecurity teams learn from each incident and adapt accordingly.

THE NEED FOR AN OFFENSIVE APPROACH IN CYDEC Another key factor revealed in the analysis is the scarcity of articles that focus on designing or implementing mechanisms based on attacking rather than defending, while in the area of counterattack, there is a study on the subject.

Most research and developments in CYDEC focus on strengthening defenses, creating more robust barriers, and more secure systems to prevent intrusions. Attackers continually evolve and improve their techniques, evading even the most advanced defenses. CYDEC, instead, features several facets in which we can carry out defense or counterattack actions to counter the opponent's advances. In fact, few articles address how to design and implement these deception systems effectively. In addition, there is an inherent reluctance in many corporate and academic environments to adopt offensive strategies. However, as threats grow more sophisticated, cybersecurity must evolve to include counterattack methods alongside protection. While the focus has been largely defensive, incorporating controlled offensive maneuvers can further disrupt attackers. Research is needed to develop ethical, legally sound strategies that use offensive deception supported by, e.g., game-theoretical models to optimize tactical decisions.

THE NEED TO ADDRESS EXPLAINABILITY, TRUSTWORTHINESS AND ROBUSTNESS OF AI IN CYDEC In the field of CYDEC, AI is emerging as a powerful tool to anticipate and counter cyberattacks. However, there is a notable gap in the scientific literature, i.e., the lack of studies addressing the explainability, trustworthiness, and robustness of AI models applied to CYDEC. This gap represents an open challenge and a significant opportunity to advance CYDEC + AI.

In this sense, explainability [207] of AI models refers to the ability to transparently understand and communicate how and why a model makes certain decisions. Currently, no articles specifically address how to make AI models in CYDEC explainable. This weakness may limit the adoption and effectiveness of these models, as users cannot be confident in automated decisions.

Moreover, trustworthiness [208] in AI models is another critical aspect that needs to be addressed. Trust is built through explainability, validation, and continuous evaluation of the models in real environments.

Last but not least, the robustness [209] of AI models in CYDEC is equally crucial. Robustness refers to the ability of a model to maintain its performance in the face of adverse conditions, such as malicious attacks or unexpected data. However, the literature lacks research that explores how to design and evaluate the robustness of AI models in the context of CYDEC, leaving a significant gap in our ability to defend against advanced threats effectively.

Ensuring that AI components are understandable, reliable, and robust under adversarial conditions is essential. Efforts in making AI-driven solutions explainable will enhance trust among cybersecurity professionals. Simultaneously, robust designs that withstand attacks will ensure these systems function effectively, thereby solidifying their role in modern CYDEC strategies.

IX. CONCLUSION AND FUTURE WORK

This article studies the evolution of CYDEC in recent years, providing the fundamentals along with an analysis of existing taxonomies, and a proposal of its own. In addition, the CYDEC frameworks in the literature are reviewed, and a generic AI-driven framework is proposed. The main characteristics of the CYDEC mechanisms proposed in the literature are analyzed, differentiating them into two groups: those that use AI and those that do not. In this context, the present work has answered the following research questions:

RQ1: What are the fundamentals aspects of CYDEC?

Section II presents the fundamental principles on which CYDEC is based. These fundamentals are divided into the solutions of the five fundamental questions: What, Where, Why, When and How. With the solution to these questions, what CYDEC is and what CYDEC can be is fully realized.

RQ2: What Cyber Deception taxonomies exist?

Section III surveys and analyzes the main taxonomies proposed by the scientific community over time. This section details the advantages and disadvantages of each of these taxonomies to propose a complete taxonomy that can efficiently classify any CYDEC mechanism.

RQ3: What are the relevant Cyber Deception frameworks?

Section IV analyzes and discusses the main frameworks and architectures that use CYDEC to defend against threats. This section examines how generic these solutions can be and the use of CYDEC for different tactics of defense or the use of AI in these frameworks. In addition, a generic framework

capable of being used in different tactics of defense with AI is proposed.

RQ4: What are the characteristics of the Cyber Deception mechanisms?

Sections VI-A and VI-B describes, analyzes, and compares the different CYDEC-based solutions in the literature. This analysis uses the previously proposed taxonomy, identifying the application scenario in which they are used and the TRL in which each solution is found and the features discussed in Section V-B. In addition, a survey of the main trends in this area is carried out.

RQ5: What are the characteristics of the AI mechanisms in Cyber Deception?

Sections VI-C and VI-D describes, analyzes, and compares the CYDEC-based solutions using AI in the literature. A similar analysis is performed in Section VI-A but now indicates which AI discipline is used. In this section, we focus on eliciting the main characteristics of AI usage for the different CYDEC solutions and identifying current trends with the help of the features in Section V-B.

RQ6: What trends, lessons learned and open challenges have emerged in Cyber Deception?

Section VIII details the main challenges we encountered in improving the different CYDEC solutions. In addition, certain gaps in the literature are identified, such as the lack of common metrics, the use of advanced AI techniques, and the lack of solutions in certain dimensions and tactics, which opens the door for future research.

As future work, it is planned to design and implement CYDEC solutions with a higher level of effectiveness and capable of simultaneously addressing several tactics of defense. To this end, applying new federated architectures and generative models can significantly improve CYDEC mechanisms, enabling collaborative learning and the creation of more creative and adaptive methods. In addition, diversifying CYDEC techniques beyond networks and improving the coordinated integration of multiple approaches will increase the difficulty for attackers. Discreteness and the forensic tactic must also be prioritized to generate valuable intelligence and strengthen defenses. Finally, implementing the framework designed in this article will be key to bridging each of the challenges addressed in this study.

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